

Equation 7 shows that the solution $\mathbf{n}_t$ can be viewed as the sum of two parts, each of which corresponds to one of the eigenvector-eigenvalue pairs of the matrix A . More generally, for an $n \times n$ matrix we have

$$(8) \quad \mathbf{n}_t = c_1 \begin{bmatrix} | \\ \mathbf{v}_1 \\ | \end{bmatrix} \lambda_1^t + c_2 \begin{bmatrix} | \\ \mathbf{v}_2 \\ | \end{bmatrix} \lambda_2^t + \cdots + c_n \begin{bmatrix} | \\ \mathbf{v}_n \\ | \end{bmatrix} \lambda_n^t$$

EXAMPLE 1 | Consider the matrix $A = \begin{bmatrix} 1 & \frac{1}{2} \\ 1 & \frac{3}{2} \end{bmatrix}$. On page 539 we found the

eigenvalues to be $\lambda_1 = 2$ and $\lambda_2 = \frac{1}{2}$, and their corresponding eigenvectors to be

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \text{ and } \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

(a) Express the solution to the recursion $\mathbf{n}_{t+1} = A\mathbf{n}_t$ in terms of these eigenvalues and eigenvectors, assuming $\mathbf{n}_0 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

(b) What happens to the vector $\mathbf{n}_t$ as $t \rightarrow \infty$?

SOLUTION

(a) To evaluate Equation 7 explicitly we need to calculate the constants c_1 and c_2 . These are determined from P^{-1} and the vector $\mathbf{n}_0$. To calculate P^{-1} we note that

$$P = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix} \text{ and therefore, using the formula for the inverse of a } 2 \times 2 \text{ matrix}$$

on page 530, we get $P^{-1} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{2}{3} & \frac{1}{3} \end{bmatrix}$. The constants c_1 and c_2 are calculated as

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = P^{-1}\mathbf{n}_0 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \text{ We obtain}$$

$$\mathbf{n}_t = (1) \begin{bmatrix} 1 \\ 2 \end{bmatrix} 2^t + (-1) \begin{bmatrix} -1 \\ 1 \end{bmatrix} \left(\frac{1}{2}\right)^t = \begin{bmatrix} 1 \\ 2 \end{bmatrix} 2^t + \begin{bmatrix} 1 \\ -1 \end{bmatrix} \left(\frac{1}{2}\right)^t$$

(b) As $t \rightarrow \infty$, the part of the solution corresponding to eigenvector $\mathbf{v}_2$ decays to zero because $\lim_{t \rightarrow \infty} \left(\frac{1}{2}\right)^t = 0$. At the same time, the part of the solution corresponding to eigenvector $\mathbf{v}_1$ grows by a factor of 2 each time step. As a result, the vector $\mathbf{n}_t$ ultimately increases in magnitude and approaches the direction of the first eigenvector,

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \text{ as shown in Figure 1.}$$

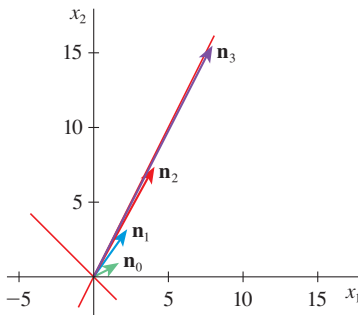


FIGURE 1

The red lines indicate all scalar multiples of the eigenvectors. The vector $\mathbf{n}_1$ was obtained by multiplying $\mathbf{n}_0$ by A . The vector $\mathbf{n}_2$ was obtained by multiplying $\mathbf{n}_1$ by A , and so on.

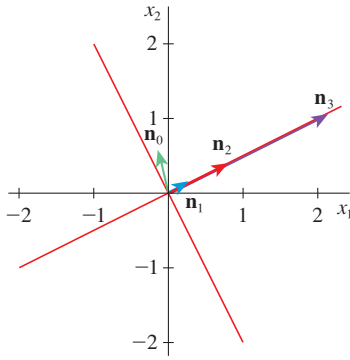
EXAMPLE 2 | In Example 8.7.2 (on page 541) we found the eigenvectors and eigenvalues of matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & \frac{1}{2} \end{bmatrix}$ to be $\lambda_1 = 0$ and $\lambda_2 = \frac{5}{2}$, and $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

(a) Express the solution to the recursion $\mathbf{n}_{t+1} = A\mathbf{n}_t$ in terms of these eigenvalues and eigenvectors, assuming $\mathbf{n}_0 = \begin{bmatrix} a \\ b \end{bmatrix}$.

(b) What happens to the vector $\mathbf{n}_t$ as $t \rightarrow \infty$?

SOLUTION

(a) We have $P = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$ and therefore $P^{-1} = \begin{bmatrix} \frac{1}{5} & -\frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} \end{bmatrix}$. Calculating $\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = P^{-1}\mathbf{n}_0$

**FIGURE 2**

The red lines indicate all scalar multiples of the eigenvectors. The vector $\mathbf{n}_1$ was obtained by multiplying $\mathbf{n}_0$ by A . The vector $\mathbf{n}_2$ was obtained by multiplying $\mathbf{n}_1$ by A , and so on.

gives the constants $c_1 = (a - 2b)/5$ and $c_2 = (2a + b)/5$. The solution is therefore

$$\mathbf{n}_t = \frac{a - 2b}{5} \begin{bmatrix} 1 \\ -2 \end{bmatrix} 0^t + \frac{2a + b}{5} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \left(\frac{5}{2}\right)^t = \frac{2a + b}{5} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \left(\frac{5}{2}\right)^t$$

Note that this solution is restricted to values of t satisfying $t \geq 1$ because 0^t is undefined when $t = 0$. The exclusion of the case $t = 0$ in the solution of matrix models is typical whenever the matrix has a zero eigenvalue.

(b) Because $\lambda_1 = 0$, the part of the solution corresponding to its eigenvector drops out of the solution. As a result, although the initial vector $\mathbf{n}_0$ can lie anywhere in the plane, $\mathbf{n}_1$ always lies on the line containing the eigenvector $\mathbf{v}_2$. The solution $\mathbf{n}_t$ lies on this axis for all subsequent times, growing in magnitude by a factor of $\frac{5}{2}$ each time step. This is illustrated in Figure 2 for the specific initial vector $\mathbf{n}_0 = [-0.15, 0.65]$. ■

EXAMPLE 3 | **BB** Class-structured population dynamics On page 520 we introduced matrix models with an example of a population of juveniles and adults that changes in size through time according to the recursion $\mathbf{n}_{t+1} = Q\mathbf{n}_t$, where

$$Q = \begin{bmatrix} 0 & 2 \\ \frac{1}{2} & \frac{1}{3} \end{bmatrix}$$

(a) Use the eigenvectors and eigenvalues of Q to find an equation that gives the number of juveniles and adults at time t for an arbitrary initial condition

$$\mathbf{n}_0 = \begin{bmatrix} j_0 \\ a_0 \end{bmatrix}$$

(b) How do the numbers of juveniles and adults change each time step once t is very large?

SOLUTION

(a) The eigenvalues of Q can be calculated as $\lambda_1 = (1 + \sqrt{37})/6 \approx 1.18$ and $\lambda_2 = (1 - \sqrt{37})/6 \approx -0.85$. After some calculation, the corresponding eigenvectors are found to be

$$\mathbf{v}_1 = \begin{bmatrix} -1 + \sqrt{37} \\ 3 \end{bmatrix} \approx \begin{bmatrix} 5.1 \\ 3 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} -1 - \sqrt{37} \\ 3 \end{bmatrix} \approx \begin{bmatrix} -7.1 \\ 3 \end{bmatrix}$$

Therefore we have

$$P = \begin{bmatrix} -1 + \sqrt{37} & -1 - \sqrt{37} \\ 3 & 3 \end{bmatrix} \quad \text{and} \quad P^{-1} = \begin{bmatrix} \frac{1}{2\sqrt{37}} & \frac{1 + \sqrt{37}}{6\sqrt{37}} \\ -\frac{1}{2\sqrt{37}} & \frac{-1 + \sqrt{37}}{6\sqrt{37}} \end{bmatrix}$$

where P^{-1} follows from the definition on page 530.

Finally, calculating $\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = P^{-1}\mathbf{n}_0$ gives the constants

$$c_1 = \frac{j_0}{2\sqrt{37}} + a_0 \left(\frac{1 + \sqrt{37}}{6\sqrt{37}} \right) \approx 0.08j_0 + 0.19a_0$$

$$c_2 = \frac{-j_0}{2\sqrt{37}} + a_0 \left(\frac{-1 + \sqrt{37}}{6\sqrt{37}} \right) \approx -0.08j_0 + 0.14a_0$$

The solution, in decimal form, is therefore

$$\mathbf{n}_t = (0.08j_0 + 0.19a_0) \begin{bmatrix} 5.1 \\ 3 \end{bmatrix} (1.18)^t + (-0.08j_0 + 0.14a_0) \begin{bmatrix} -7.1 \\ 3 \end{bmatrix} (-0.85)^t$$

(b) As t increases, the part of the solution corresponding to the first eigenvector grows by a factor $\lambda_1 \approx 1.18$ each time step. The part of the solution corresponding to the second eigenvector alternates in sign and decays in magnitude each time step, asymptotically going to zero. Therefore, for any choice of initial vector $\mathbf{n}_0$, once t gets large the solution vector $\mathbf{n}_t$ will grow in magnitude by a factor ≈ 1.18 each time step and asymptotically approach the line defined by the eigenvector

$$\mathbf{v}_1 = \begin{bmatrix} -1 + \sqrt{37} \\ 3 \end{bmatrix} \approx \begin{bmatrix} 5.1 \\ 3 \end{bmatrix}$$

Figure 3 illustrates the specific case where $\mathbf{n}_0 = [0.15, 0.65]$ (measured in thousands of individuals).

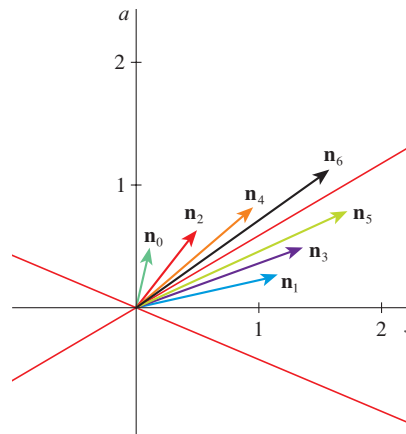


FIGURE 3

The red lines indicate all scalar multiples of the eigenvectors. The vector $\mathbf{n}_1$ was obtained by multiplying $\mathbf{n}_0$ by A . The vector $\mathbf{n}_2$ was obtained by multiplying $\mathbf{n}_1$ by A , and so on.

Thus, once t is large, both juvenile and adult subpopulations grow by a factor 1.18 each time step [see Figure 4(a)]. Their relative abundances approach a constant ratio given by the ratio of the components of $\mathbf{v}_1$. This is illustrated in Figure 4(b).

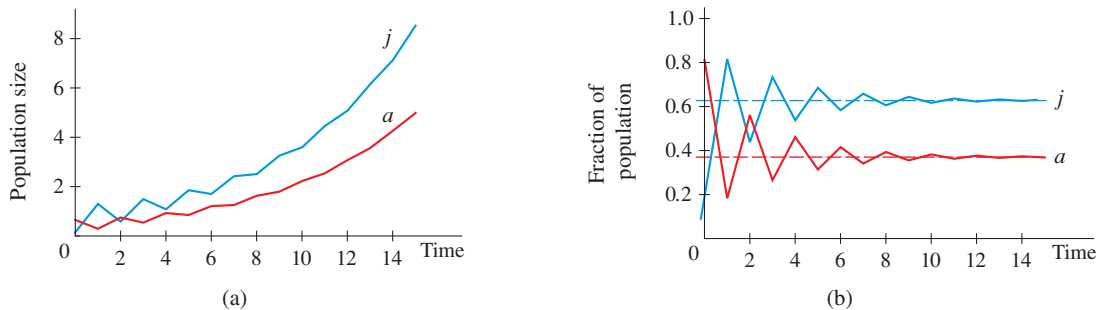


FIGURE 4 Plots of the components of vectors in Figure 3 as functions of time. Dashed lines in part (b) indicate the components of the vector $k\mathbf{v}_1$, where $k = 1/(5.1 + 3) = 1/8.1$. This scaling of $\mathbf{v}_1$ was chosen so that the components of the resulting vector add to 1.

■ Solutions with Complex Eigenvalues

In Section 8.7 we saw that the eigenvector-eigenvalue pairs of some matrices involve complex numbers. How do we interpret Equation 8 in this case? An important result from complex analysis called De Moivre's Theorem provides the key.

Complex numbers and De Moivre's Theorem are discussed in Appendix G.

De Moivre's Theorem provides a relationship between powers of complex numbers and the trigonometric functions sine and cosine. If $a + bi$ is a complex number (where a and b are real numbers) and k is a positive integer, then

$$(a + bi)^k = r^k(\cos k\theta + i \sin k\theta)$$

where $r = \sqrt{a^2 + b^2}$ is called the modulus of $a + bi$ and θ is called its argument. Specifically, $\theta = \tan^{-1}(b/a)$.

To illustrate the consequences of De Moivre's Theorem, we return to Example 8.7.3.

There we studied the matrix $A = \begin{bmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{bmatrix}$ and found its eigenvalues to be $\lambda_1 = \sqrt{3} + i$ and $\lambda_2 = \sqrt{3} - i$, with associated eigenvectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -i \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 1 \\ i \end{bmatrix}$. Substituting these into Equation 7 gives the solution

$$\begin{aligned} \mathbf{n}_t &= c_1 \begin{bmatrix} 1 \\ -i \end{bmatrix} (\sqrt{3} + i)^t + c_2 \begin{bmatrix} 1 \\ i \end{bmatrix} (\sqrt{3} - i)^t \\ &= c_1 \begin{bmatrix} 1 \\ -i \end{bmatrix} 2^t \left(\cos\left(\frac{\pi}{6}t\right) + i \sin\left(\frac{\pi}{6}t\right) \right) + c_2 \begin{bmatrix} 1 \\ i \end{bmatrix} 2^t \left(\cos\left(\frac{\pi}{6}t\right) - i \sin\left(\frac{\pi}{6}t\right) \right) \end{aligned}$$

where we have used the facts that

$$r = \sqrt{(\sqrt{3})^2 + 1^2} = 2 \quad \theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6} \quad \theta = \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6}$$

as well as the fact that sine is an odd function.

The final step is to calculate the constants c_1 and c_2 . First, we have $P = \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix}$ and

$$\text{therefore } P^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{i}{2} \\ \frac{1}{2} & -\frac{i}{2} \end{bmatrix}. \text{ The constants } c_1 \text{ and } c_2 \text{ are then}$$

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = P^{-1}\mathbf{n}_0 = \begin{bmatrix} (x_0 + y_0i)/2 \\ (x_0 - y_0i)/2 \end{bmatrix}$$

Substituting these into the solution gives

$$\begin{aligned} \mathbf{n}_t &= \frac{x_0 + y_0i}{2} \begin{bmatrix} 1 \\ -i \end{bmatrix} 2^t \left(\cos\left(\frac{\pi}{6}t\right) + i \sin\left(\frac{\pi}{6}t\right) \right) + \frac{x_0 - y_0i}{2} \begin{bmatrix} 1 \\ i \end{bmatrix} 2^t \left(\cos\left(\frac{\pi}{6}t\right) - i \sin\left(\frac{\pi}{6}t\right) \right) \\ &= \frac{1}{2} \begin{bmatrix} x_0 + y_0i \\ -x_0i + y_0 \end{bmatrix} 2^t \left(\cos\left(\frac{\pi}{6}t\right) + i \sin\left(\frac{\pi}{6}t\right) \right) + \frac{1}{2} \begin{bmatrix} x_0 - y_0i \\ x_0i + y_0 \end{bmatrix} 2^t \left(\cos\left(\frac{\pi}{6}t\right) - i \sin\left(\frac{\pi}{6}t\right) \right) \\ &= 2^t \begin{bmatrix} x_0 \cos\left(\frac{\pi}{6}t\right) - y_0 \sin\left(\frac{\pi}{6}t\right) \\ x_0 \sin\left(\frac{\pi}{6}t\right) + y_0 \cos\left(\frac{\pi}{6}t\right) \end{bmatrix} = 2^t \begin{bmatrix} \cos\left(\frac{\pi}{6}t\right) & -\sin\left(\frac{\pi}{6}t\right) \\ \sin\left(\frac{\pi}{6}t\right) & \cos\left(\frac{\pi}{6}t\right) \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} \end{aligned}$$

The second-last equality follows after considerable simplification, in which all of the imaginary parts of the solution cancel. As conjectured in Example 8.7.3, we can now see, using the result of Exercise 8.7.6, that $\mathbf{n}_t$ gets stretched by a factor of 2 each time step and rotated through an angle of $\pi/6$ radians. Figure 5 illustrates a specific example where $\mathbf{n}_0 = [0.35, -0.25]$. The components of $\mathbf{n}_t$ from Figure 5 are plotted as functions of time in Figure 6.

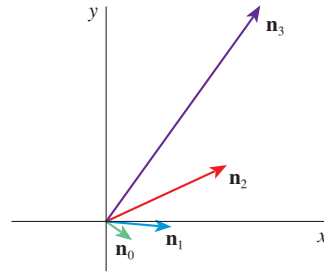


FIGURE 5

The vector $\mathbf{n}_1$ was obtained by multiplying $\mathbf{n}_0$ by A . The vector $\mathbf{n}_2$ was obtained by multiplying $\mathbf{n}_1$ by A , and so on.

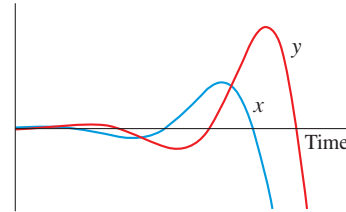


FIGURE 6

The components of $\mathbf{n}_t$ as functions of time

The cancellation of imaginary parts in the solution of the preceding example occurs in general when the eigenvalues of A are complex. The solution in such cases always consists of the oscillatory functions sine and cosine, and the solution grows or decays in magnitude depending on the modulus of the eigenvalues (see Exercise 29).

■ Perron-Frobenius Theory

In Examples 1–3 the long-term behavior of the solution (that is, the behavior of $\mathbf{n}_t$ as $t \rightarrow \infty$) was determined by just one of the eigenvector-eigenvalue pairs of the matrix A . The reason can be seen from Equation 8: Suppose that the eigenvalues in Equation 8 are ordered in terms of their absolute value (or modulus, if they are complex) from the largest λ_1 to the smallest λ_n , with $|\lambda_1| > |\lambda_i|$ for all $i \neq 1$ (that is, λ_1 is strictly larger in modulus than all other eigenvalues). Dividing both sides of Equation 8 by λ_1^t gives

$$\frac{\mathbf{n}_t}{\lambda_1^t} = c_1 \begin{bmatrix} | \\ \mathbf{v}_1 \\ | \end{bmatrix} + c_2 \begin{bmatrix} | \\ \mathbf{v}_2 \\ | \end{bmatrix} \frac{\lambda_2^t}{\lambda_1^t} + \cdots + c_n \begin{bmatrix} | \\ \mathbf{v}_n \\ | \end{bmatrix} \frac{\lambda_n^t}{\lambda_1^t}$$

Now taking the limit $t \rightarrow \infty$, we have

$$\lim_{t \rightarrow \infty} \frac{\mathbf{n}_t}{\lambda_1^t} = c_1 \begin{bmatrix} | \\ \mathbf{v}_1 \\ | \end{bmatrix}$$

This shows that, in the long term, as $t \rightarrow \infty$ the components of $\mathbf{n}_t$ become proportional to the components of the eigenvector $\mathbf{v}_1$. Thus the solution vector asymptotically approaches the line containing $\mathbf{v}_1$, regardless of the initial vector $\mathbf{n}_0$. This is a powerful piece of information: In the long run, all components of the vector $\mathbf{n}_t$ change by a factor λ_1 each time step, with the relative sizes of its components being proportional to the components of $\mathbf{v}_1$.

The preceding considerations are true only when λ_1 is strictly larger in magnitude than the other eigenvalues, and when the components of the associated eigenvector are positive. The power of matrix models in the life sciences stems, in large part, from the fact that many biological processes correspond to matrices whose eigenvectors and eigenvalues satisfy these conditions. The following definition and theorem specify precisely the kinds of matrices that have these properties.

Definition An $n \times n$ matrix A whose entries are nonnegative is called **primitive** if there exists a positive integer k such that all the entries of A^k are positive.

With this definition, we can now state the following theorem.

Perron-Frobenius Theorem Suppose that A is an $n \times n$ matrix whose entries are all nonnegative. If A is primitive, then all of the following are true:

1. There exists an eigenvalue of A , call it λ_1 , that is real and positive.
2. $|\lambda_1| > |\lambda_i|$ for all $i \neq 1$ (that is, λ_1 is greater in magnitude than all other eigenvalues).
3. The components of the eigenvector associated with λ_1 are all positive.

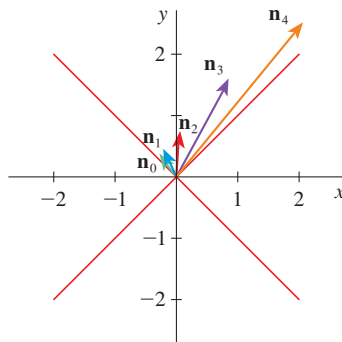


FIGURE 7
The red lines indicate all scalar multiples of the eigenvectors. Each vector was obtained from the preceding one through multiplication by A .

EXAMPLE 4 | Consider the recursion $\mathbf{n}_{t+1} = A\mathbf{n}_t$, where $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.
(a) What happens to the magnitude of $\mathbf{n}_t$ as $t \rightarrow \infty$?
(b) What happens to the direction of $\mathbf{n}_t$ as $t \rightarrow \infty$?

SOLUTION Since A is primitive, we can apply the Perron-Frobenius Theorem. One eigenvalue is real and positive, larger in modulus than the other, and has an eigenvector with positive components. In this case we can calculate them explicitly, obtaining

$$\lambda_1 = 3 \text{ and } \lambda_2 = 1, \text{ along with the eigenvectors } \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ and } \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

- (a) As $t \rightarrow \infty$ the vector $\mathbf{n}_t$ eventually grows in magnitude by a factor of $\lambda_1 = 3$ each time step, regardless of the initial vector $\mathbf{n}_0$.
(b) As $t \rightarrow \infty$ the vector $\mathbf{n}_t$ eventually points in the direction of $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, regardless of the initial vector $\mathbf{n}_0$. Figure 7 illustrates this behavior when $\mathbf{n}_0 = [-0.35, 0.45]$. ■

EXAMPLE 5 | **BB** Age structure and the Leslie matrix Consider an age-structured population whose size is governed by the equation

$$\begin{bmatrix} n_{1,t+1} \\ n_{2,t+1} \\ n_{3,t+1} \end{bmatrix} = \begin{bmatrix} 2 & 2 & 2 \\ \frac{1}{10} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} n_{1,t} \\ n_{2,t} \\ n_{3,t} \end{bmatrix}$$

What is the long-term fate of the population?

SOLUTION Denoting the Leslie matrix in the equation by L , we begin by exploring the first few powers of L . We find that L^3 has entries that are all positive, meaning that L is primitive. Therefore we can apply the Perron-Frobenius Theorem and so there is a positive, real eigenvalue that is larger in magnitude than all others. In the long term the

population will grow or decay depending on whether this eigenvalue is larger or smaller than 1.

Calculating the characteristic polynomial of L gives $f(\lambda) = \lambda^3 - 2\lambda^2 - \frac{1}{5}\lambda - \frac{1}{10}$. This polynomial does not factor and therefore its roots are difficult to find. But the Perron-Frobenius Theorem tells us that the largest root of the equation $f(\lambda) = 0$ is positive. Therefore, since $\lim_{\lambda \rightarrow \infty} f(\lambda) = \infty$, as the value of λ increases, the graph of the function $f(\lambda)$ crosses the horizontal axis for the last time in one of the two ways illustrated in Figure 8.

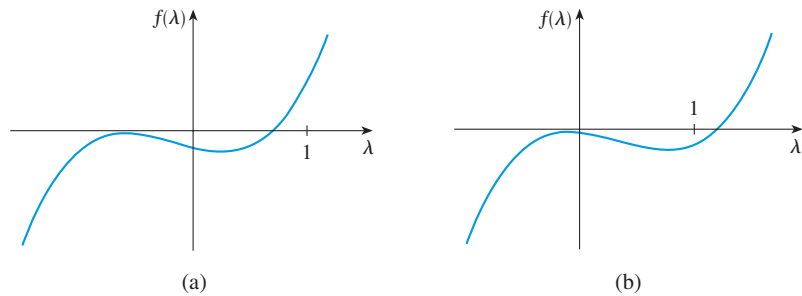


FIGURE 8

Furthermore, by noting that $f(1) = -13/10 < 0$, we can see that Figure 8(a) cannot be correct because it requires $f(1) > 0$. Indeed, because $f(1) = -13/10$ and $\lim_{\lambda \rightarrow \infty} f(\lambda) = \infty$, the Intermediate Value Theorem tells us that the largest root in the Perron-Frobenius Theorem must be larger than 1. Consequently, in the long term the population will grow. The Perron-Frobenius Theorem also tells us that the fraction of the population made up of each age class stabilizes to constant values that are proportional to the components of the eigenvector associated with λ_1 . Figure 9 illustrates the case where $n_{1,0} = 0.1$, $n_{2,0} = 1$, and $n_{3,0} = 1$.

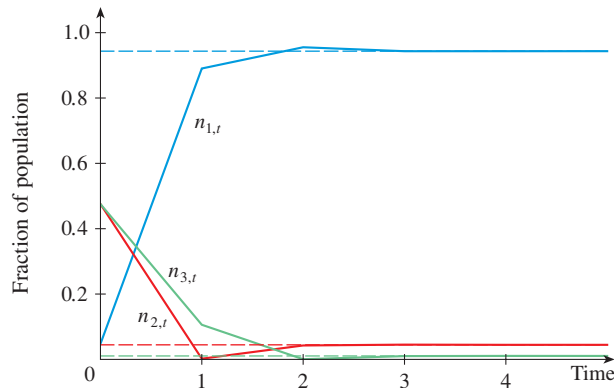


FIGURE 9

Dashed lines indicate the components of the eigenvector associated with the largest eigenvalue.

EXERCISES 8.8

1–8 Show that $A = PDP^{-1}$, where P is a matrix whose columns are the eigenvectors of A , and D is a diagonal matrix with the corresponding eigenvalues.

1. $A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$

2. $A = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix}$

3. $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

4. $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

5. $A = \begin{bmatrix} 1 & 2 \\ -3 & 3 \end{bmatrix}$

6. $A = \begin{bmatrix} 0 & a \\ 0 & b \end{bmatrix}$ with $b \neq 0$

CAS 7. $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

CAS 8. $A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}$

9–14 Show that $A^2 = PD^2P^{-1}$, where P is a matrix whose columns are the eigenvectors of A , and D is a diagonal matrix with the corresponding eigenvalues.

9. $A = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$

10. $A = \begin{bmatrix} 1 & 4 \\ 0 & 4 \end{bmatrix}$

11. $A = \begin{bmatrix} 2 & 2 \\ 0 & 1 \end{bmatrix}$

12. $A = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$

13. $A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ with $a \neq b$

14. $A = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$ with $a \neq c$

15–20 Express the solution to the recursion $\mathbf{n}_{t+1} = A\mathbf{n}_t$ in terms of the eigenvectors and eigenvalues of A , assuming that

$\mathbf{n}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

15. $A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$

16. $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

17. $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$

18. $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

19. $A = \begin{bmatrix} 1 & a \\ 0 & 2 \end{bmatrix}$ with $a \neq 0$

20. $A = \begin{bmatrix} 1 & a \\ 0 & b \end{bmatrix}$ with $a \neq 0$ and $b \neq 1$

21–25 Express the solution to the recursion $\mathbf{n}_{t+1} = A\mathbf{n}_t$ in terms of the eigenvectors and eigenvalues of A , assuming arbitrary initial conditions.

21. $A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ with $a \neq b$ 22. $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

23. $A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$

24. $A = \begin{bmatrix} a & 1 \\ 0 & 1 \end{bmatrix}$ with $a \neq 1$

25. $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ with $b \neq 0$

26–29 Like scalars, vectors with complex components can be broken into real and imaginary parts. For example, $\mathbf{v} = \mathbf{a} + bi$, where $\mathbf{a}$ and $\mathbf{b}$ are the real and imaginary parts of the vector, respectively. The complex conjugate of $\mathbf{v}$ is $\bar{\mathbf{v}} = \mathbf{a} - bi$. The analogous notation holds for matrices. The following rules from complex analysis, familiar from the scalar case, carry over to vectors and matrices: $\overline{r\mathbf{x}} = \bar{r}\bar{\mathbf{x}}$, $\overline{B\mathbf{x}} = \bar{B}\bar{\mathbf{x}}$, $\overline{BC} = \bar{B}\bar{C}$, and $\overline{rB} = \bar{r}\bar{B}$, where r is a scalar, $\mathbf{x}$ is a vector, and B and C are matrices. (Note: Complex numbers are discussed in Appendix G.)

26. Suppose that A is a matrix with real entries. Show that if λ is a complex eigenvalue of A , then its associated eigenvector will also be complex.

27. Suppose that A is a matrix with real entries. If λ is a complex eigenvalue of A with associated eigenvector $\mathbf{x}$, show that $\bar{\lambda}$ is also an eigenvalue of A with associated eigenvector $\bar{\mathbf{x}}$ (that is, eigenvalues and eigenvectors of real matrices always come in complex conjugate pairs).

28. Suppose that $T = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ with $a \neq 0$ and $b \neq 0$.

- (a) Show that the eigenvalues of T are $\lambda = a \pm bi$.
 (b) Show that T can be written as

$$T = r \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

where $r = \sqrt{a^2 + b^2}$ is the modulus of the eigenvalues and θ is the argument. This shows that the matrix T represents a rotation that also scales all vectors by a factor r . (Note: Recall Exercise 8.7.6.)

29. Suppose that A is any 2×2 matrix with complex eigenvalues $\lambda = a \pm bi$. We can write $A = PDP^{-1}$, where P is a matrix whose columns contain the (complex) eigenvectors of A , and D is a diagonal matrix with their associated (complex) eigenvalues.

(a) Show that $D = R^{-1}TR$, where

$$R = \begin{bmatrix} i & -i \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad T = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

- (b) Part (a) shows that A can be written as $A = PR^{-1}TRP^{-1}$. Show that $S = PR^{-1}$ is a 2×2 matrix whose columns are the real and imaginary components of the eigenvectors of A .
 (c) Part (b) shows that $A = STS^{-1}$, where S is a 2×2 matrix whose columns are the real and imaginary components of the eigenvectors of A . In light of Exercise 28, show that we can then write A as

$$A = rS \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} S^{-1}$$

where $r = \sqrt{a^2 + b^2}$ is the modulus of the eigenvalues and θ is the argument.

(d) Show that

$$A^t = r^t S \begin{bmatrix} \cos \theta t & -\sin \theta t \\ \sin \theta t & \cos \theta t \end{bmatrix} S^{-1}$$

This proves that all 2×2 matrices with complex eigenvalues rotate vectors by an angle θ and stretch them by a factor r each time step.

30–34 Express the solution to the recursion $\mathbf{n}_{t+1} = A\mathbf{n}_t$ in terms of the eigenvectors and eigenvalues of A . Use De Moivre's Theorem to simplify the solution if appropriate.

30. $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ with $\mathbf{n}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

31. $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ with $\mathbf{n}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

32. $A = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$ with $\mathbf{n}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

33. $A = \begin{bmatrix} 1 & 5 \\ -1 & 1 \end{bmatrix}$ with $\mathbf{n}_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

34. $A = \begin{bmatrix} 1 & -a \\ a & 1 \end{bmatrix}$ with $\mathbf{n}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ where a is a real number

35. Vectorcardiography In Exercise 8.7.26 we modeled the heart voltage vector with the recursion $\mathbf{v}_{t+1} = A\mathbf{v}_t$, where

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

(a) Can you apply the Perron-Frobenius Theorem to this model?

(b) Using the initial condition $\mathbf{v}_0 = \begin{bmatrix} 0.3 \\ -0.2 \end{bmatrix}$, express the solution to the recursion in terms of the eigenvectors and eigenvalues of A .

(c) Describe the long-term behavior of $\mathbf{v}_t$.

36. Antigenic evolution In Exercise 8.7.27 we modeled antigenic evolution with the recursion $\mathbf{x}_{t+1} = A\mathbf{x}_t$, where

$$A = \begin{bmatrix} 2 & 3 \\ 0 & \frac{9}{10} \end{bmatrix}$$

(a) Using the initial condition $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, express the solution to the recursion in terms of the eigenvectors and eigenvalues of A .

(b) Describe the long-term behavior of $\mathbf{x}_t$.

37. Mutation In Exercise 8.6.34 we modeled mutation with a recursion of the form $\mathbf{y}_{t+1} = A\mathbf{y}_t$. Suppose

$$A = \begin{bmatrix} \frac{9}{10} & \frac{1}{5} \\ \frac{1}{10} & \frac{4}{5} \end{bmatrix}$$

(a) Calculate the eigenvalues of A .

(b) In light of the Perron-Frobenius Theorem, what is the long-term behavior of $\mathbf{y}_t$?

(c) Using the initial condition $\mathbf{y}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, express the solution to the recursion in terms of the eigenvectors and eigenvalues of A .

(d) Verify your answer to part (b) using the solution obtained in part (c).

38. Leslie matrices In Exercise 8.7.29 we modeled an age-structured population using the recursion $\mathbf{n}_{t+1} = L\mathbf{n}_t$, where

$$L = \begin{bmatrix} b & 2 \\ \frac{1}{2} & 0 \end{bmatrix} \quad b > 0$$

(a) Verify that the Perron-Frobenius Theorem can be applied to this model.

(b) Calculate the eigenvalues of L .

(c) Given your answers to parts (a) and (b), what is the long-term behavior of $\mathbf{n}_t$ as a function of b ?

(d) Using the initial condition $\mathbf{n}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, express the solution to the recursion in terms of the eigenvectors and eigenvalues of L .

(e) Verify your answer to part (c) using the solution you obtained in part (d).

39. Methylation In Exercise 8.5.20 we modeled DNA methylation with the recursion $\mathbf{x}_{t+1} = A\mathbf{x}_t$, where

$$A = \begin{bmatrix} \frac{8}{10} & \frac{1}{2} \\ \frac{2}{10} & \frac{1}{2} \end{bmatrix}$$

(a) Calculate the eigenvalues of A .

(b) The Perron-Frobenius Theorem can clearly be applied to the matrix A . Given your answer to part (a), what is the long-term behavior of $\mathbf{x}_t$?

(c) Using the initial condition $\mathbf{x}_0 = \begin{bmatrix} a \\ 1 - a \end{bmatrix}$, express the solution to the recursion in terms of the eigenvectors and eigenvalues of A .

(d) Verify your answer to part (b) using the solution you obtained in part (c).

- CAS 40. Breast cancer** In Example 8.5.2 a model for the dynamics of breast cancer was given by the recursion $\mathbf{n}_{t+1} = A\mathbf{n}_t$, where

$$A = \begin{bmatrix} 0.998 & 0.45 & 0 \\ 0.002 & 0.1 & 0 \\ 0 & 0.45 & 1 \end{bmatrix}$$

- (a) Using the initial condition $\mathbf{n}_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, express the solution to the recursion in terms of the eigenvectors and eigenvalues of A .
- (b) Describe the long-term behavior of $\mathbf{n}_t$.

- 41. Fibonacci numbers** The n th Fibonacci number is generated by the recursion equation $F_n = F_{n-1} + F_{n-2}$ with $F_0 = F_1 = 1$. Define the two new variables $x_n = F_n$ and $y_n = F_{n-1}$.

- (a) Show that the vector $\mathbf{z}_n = \begin{bmatrix} x_n \\ y_n \end{bmatrix}$ obeys the recursion

$$\mathbf{z}_n = A\mathbf{z}_{n-1}, \text{ where } A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}.$$

- (b) Find the eigenvalues of A .
- (c) Can the Perron-Frobenius Theorem be applied to the matrix A ? If so, what is the long-term behavior of the vector $\mathbf{z}_n$?
- (d) Find a formula for the n th Fibonacci number.

PROJECT The Emergence of Geometric Order in Proliferating Cells

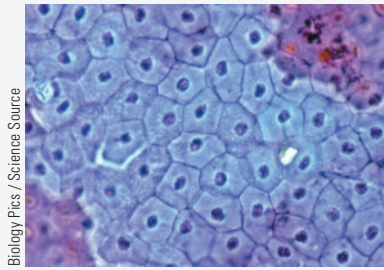


FIGURE 1

When some cells divide they form thin sheets in which the majority of cells are each adjacent to six other cells. This gives the sheet a hexagonal pattern as shown in Figure 1. In this project we will see how a mathematical model¹ predicts the emergence of this pattern simply by the way cells divide.

Each cell in a sheet can be viewed as a polygon having a certain number of edges and vertices (see Figure 2). We assume that each cell has a minimum of four edges and that, when a cell division occurs (that is, when a cell splits into two daughter cells), the line of cell division always connects two edges, subdividing each edge (see Figure 3). The cells divide asynchronously and we take a single time step to be the time after which all cells in the sheet have divided once. Let c_t , v_t , and e_t denote the total number of cells, vertices, and edges after t cell divisions. Since each cell divides into two in a single time step, we have $c_{t+1} = 2c_t$.

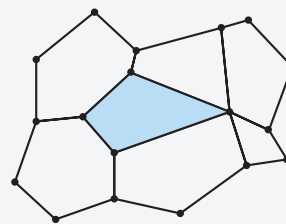


FIGURE 2

The shaded cell has four vertices and edges.

1. Explain why the total number of vertices in the cell sheet obeys the recursion $v_{t+1} = v_t + 2c_t$, and why the total number of edges obeys the recursion $e_{t+1} = e_t + 3c_t$. (Figure 3 might be helpful for this.)

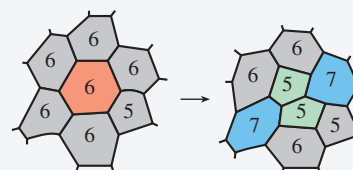


FIGURE 3

Numerals in each cell indicate the number of edges.

1. M. Gibson et al., "The Emergence of Geometric Order in Proliferating Metazoan Epithelia," *Nature* 442 (2006): 1038–41.

2. Write the system of recursion equations for c_t , v_t , and e_t in matrix notation.
3. What are the eigenvalues of the matrix from Problem 2? Show that $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, and $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ are eigenvectors corresponding to these eigenvalues.
4. Write the solution to the recursion from Problem 2 in terms of its eigenvectors and eigenvalues for an arbitrary initial condition.
5. If we ignore complications arising from the fact that cells on the boundary are not completely surrounded by other cells, then the average number of sides per cell at time t is $s_t = 2e_t/c_t$. Why?
6. Show that, in the long term (that is, when $t \rightarrow \infty$), the average number of sides per cell s_t approaches six.

Chapter 8 REVIEW

CONCEPT CHECK

1. What is the difference between a vector and a scalar?
2. How do you add two vectors geometrically? How do you add them algebraically?
3. If $\mathbf{a}$ is a vector and c is a scalar, how is $c\mathbf{a}$ related to $\mathbf{a}$ geometrically? How do you find $c\mathbf{a}$ algebraically?
4. How do you find the vector from one point to another algebraically?
5. How do you find the dot product $\mathbf{a} \cdot \mathbf{b}$ of two vectors if you know their lengths and the angle between them? What if you know their components?
6. How are dot products useful?
7. Write expressions for the scalar and vector projections of $\mathbf{b}$ onto $\mathbf{a}$. Illustrate with diagrams.
8. What is the equation of a sphere?
9. (a) How do you tell if two vectors are parallel?
(b) How do you tell if two vectors are perpendicular?
10. What is a symmetric matrix?
11. If a matrix A rotates vectors counterclockwise by θ degrees, what does the matrix A^{-1} do?
12. If a 2×2 matrix has complex eigenvalues, what does this matrix do to vectors upon multiplication?
13. Suppose A is a matrix and k is a positive integer. What does the notation A^k mean?
14. What is the relationship between the inverse of a matrix and the determinant of a matrix?
15. Explain what eigenvalues and eigenvectors are.
16. Why does a 2×2 matrix have two eigenvalues?
17. Suppose a 2×2 matrix with real entries has complex eigenvalues. Why must the eigenvalues be complex conjugates?
18. Why is it sometimes useful to write a matrix A in the form $A = PDP^{-1}$, where D is a diagonal matrix?
19. What does the Perron-Frobenius Theorem say, and why is it useful?

Answers to the Concept Check can be found on the back endpapers.

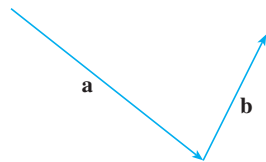
TRUE-FALSE QUIZ

Determine whether the statement is true or false. If it is true, explain why. If it is false, explain why or give an example that disproves the statement.

- For any vectors $\mathbf{u}$ and $\mathbf{v}$ in V_3 , $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$.
- For any scalar c and vectors $\mathbf{u}$ and $\mathbf{v}$, $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$.
- If $\mathbf{u} \cdot \mathbf{v} = 0$, then $\mathbf{u} = \mathbf{0}$ or $\mathbf{v} = \mathbf{0}$.
- If $\mathbf{u} = [u_1, u_2]$ and $\mathbf{v} = [v_1, v_2]$, then $\mathbf{u} \cdot \mathbf{v} = [u_1v_1, u_2v_2]$.
- The set of points $\{(x, y, z) \mid x^2 + y^2 = 1\}$ is a circle.
- The dot product of parallel vectors is zero.
- For any square matrices of the same size, $AB = BA$.
- For any square matrices of the same size, $A(B + C) = AB + AC$.
- The determinant of the identity matrix is 1.
- If $A\mathbf{x} = a\mathbf{x}$ for a square matrix A , vector $\mathbf{x}$, and scalar a , where $\mathbf{x} \neq \mathbf{0}$, then a is an eigenvalue of A .
- If $\mathbf{y}$ is an eigenvector of A , then $A\mathbf{y} = k\mathbf{y}$ for some scalar k .
- If a square matrix A is singular, it has a zero eigenvalue.
- The eigenvalues of any square matrix lie on its diagonal.
- If a 2×2 matrix A has complex eigenvalues, then it stretches and rotates vectors.
- All square matrices A with distinct eigenvalues can be written as $A = PDP^{-1}$.
- The inhomogeneous system of equations $A\mathbf{x} = \mathbf{b}$, where $\mathbf{b} \neq \mathbf{0}$, has either the trivial solution or an infinite number of solutions.
- The homogeneous system of equations $A\mathbf{x} = \mathbf{0}$ has either the trivial solution or an infinite number of solutions.
- If $\mathbf{u}$ and $\mathbf{v}$ are in V_3 , then $|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$.

EXERCISES

- Find an equation of the sphere that passes through the point $(6, -2, 3)$ and has center $(-1, 2, 1)$.
 - Find the curve in which this sphere intersects the yz -plane.
 - Find the center and radius of the sphere
$$x^2 + y^2 + z^2 - 8x + 2y + 6z + 1 = 0$$
- Copy the vectors in the figure and use them to draw each of the following vectors.
 - $\mathbf{a} + \mathbf{b}$
 - $\mathbf{a} - \mathbf{b}$
 - $-\frac{1}{2}\mathbf{a}$
 - $2\mathbf{a} + \mathbf{b}$



- A hypersphere with center at point P and radius r in $\mathbb{R}^n$ is the set of all points that are a distance r from P . Give the equation of a hypersphere.
- Calculate the given quantity if

$$\mathbf{a} = [1, 1, -2]$$

$$\mathbf{b} = [3, -2, 1]$$

$$\mathbf{c} = [0, 1, -5]$$
 - $2\mathbf{a} + 3\mathbf{b}$
 - $|\mathbf{b}|$
 - $\mathbf{a} \cdot \mathbf{b}$
 - $\text{comp}_{\mathbf{a}}\mathbf{b}$

- $\text{proj}_{\mathbf{a}}\mathbf{b}$
- The angle between $\mathbf{a}$ and $\mathbf{b}$ (correct to the nearest degree)

- Find the values of x such that the vectors $[3, 2, x]$ and $[2x, 4, x]$ are orthogonal.
- Find two unit vectors that are orthogonal to both $[0, 1, 2]$ and $[1, -2, 3]$.
- Find the acute angle between the lines.
 - $2x - y = 3, \quad 3x + y = 7$
 - $x + 2y = 7, \quad 5x - y = 2$
- Find the acute angle between two diagonals of a cube.
- Use a scalar projection to show that the distance from a point $P_1(x_1, y_1)$ to the line $ax + by + c = 0$ is

$$\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Use this formula to find the distance from the point $(-2, 3)$ to the line $3x - 4y + 5 = 0$.

- Viral identification** Viruses from the same geographic location tend to cluster together in antigenic space. Suppose all viruses from Asia fall within a sphere of radius 1 centered at $(2, 3, 1)$ in antigenic space, viruses from North America fall within a sphere of radius 1 centered at $(-3, 2, 0)$, and those from Europe fall within a sphere of radius 1 centered at $(1, 0, 1)$. (All numbers are dimensionless quantities.) A patient is infected with a virus located at

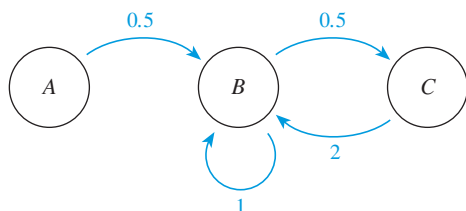
(2.5, 3, 1.75) in antigenic space. In which geographic location was this patient infected?

- 11. Genome expression divergence** The genome expression profiles of two closely related species are given by the points $(-1, 3)$ and $(1, 1)$ in dimensionless quantities.

- Calculate the difference in expression profile as measured by distance in “expression space.”
- Suppose the points in expression space are treated as the tips of position vectors. Calculate the difference in expression profile as measured by the angle between the expression vectors.

- 12. Influenza** Each month all individuals in a city are characterized as being either susceptible to influenza infection, currently infected, or resistant to infection. Suppose that susceptible individuals can become infected, infected individuals can either become susceptible again or resistant, and resistant individuals remain resistant. Draw a matrix diagram for this situation.

- 13.** Construct the matrix model for the following matrix diagram.



- 14. Leslie matrix modeling** Suppose that a species lives for a maximum of three years and individuals of all age classes produce three new offspring each year. Further, suppose the annual survival rates are 10% at age 1 and 30% at age 2.

- Construct the matrix diagram for this situation.
- Construct the matrix model from the diagram in part (a).

- 15.** Calculate the given quantity, if it is defined, using the given matrices:

$$A = \begin{bmatrix} 7 & 0 \\ 1 & 9 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 4 \\ 1 & 0 \\ 0 & 9 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 2 \\ 9 & 6 \end{bmatrix}$$

$$D = \begin{bmatrix} 15 & 0 & 7 \\ 0 & 5 & 2 \end{bmatrix}$$

- $A - D$
- $2C + A$
- $A + 2B + C$
- $D - B$
- $B^T + D$
- $2C - A^{-1}$

- 16.** Calculate the given quantity, if it is defined, using the given matrices:

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 9 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 1 \\ 1 & 9 \\ 0 & 9 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 2 \\ 9 & 2 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 1 & 7 \\ 0 & 3 & 2 \end{bmatrix}$$

- AD
- $C + C^T$
- $A(B + C)$
- $(DB)^{-1}$
- $(C - 2A)B$
- $ABCD$

- 17.** For all matrices $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ where all components a_{ij} are nonzero, show that if $A = A^{-1}$, then $\det A = -1$ and $a_{11} = -a_{22}$.

- 18.** If A is nonsingular, then $\det A^{-1} = 1/(\det A)$. Verify this theorem for 2×2 matrices.

- 19.** If I is the $n \times n$ identity matrix, then $\det I = 1$. Verify this theorem for 2×2 matrices.

- 20. Stage-structured population** Suppose a population contains juveniles and adults, and the vector $\mathbf{n}_t$ of the numbers of each of these changes from one year to the next according to the equation

$$\mathbf{n}_{t+1} = \begin{bmatrix} 0 & 2 \\ \frac{1}{2} & \frac{1}{3} \end{bmatrix} \mathbf{n}_t$$

If the number of juveniles and adults in the current year is given by the vector $\begin{bmatrix} 30 \\ 8 \end{bmatrix}$, find the numbers of juveniles and adults in the previous year.

21–24 Solve the system of equations.

$$\begin{array}{ll} 21. & 3x - y = 2 \\ & x + 7y = 4 \end{array} \qquad \begin{array}{ll} 22. & 2x + y = 2 \\ & 4x - y = 4 \end{array}$$

$$\begin{array}{ll} 23. & 7x + y = 0 \\ & 14x + 2y = 0 \end{array} \qquad \begin{array}{ll} 24. & x - 2y = 2 \\ & 3x - 6y = 4 \end{array}$$

- 25. Leslie matrix** Consider the following model for an age-structured population with three age classes:

$$\mathbf{n}_{t+1} = \begin{bmatrix} 1 & 2 & 4 \\ \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \end{bmatrix} \mathbf{n}_t$$

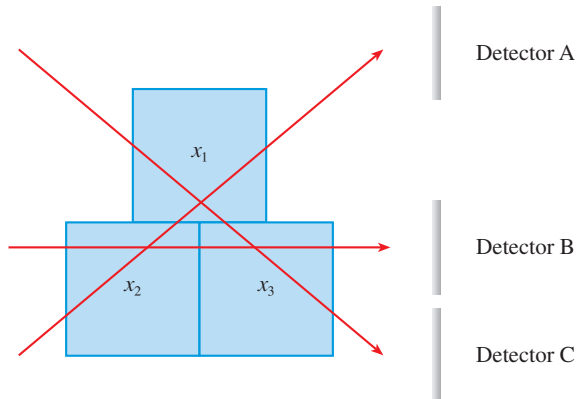
An equilibrium is a value of the vector for which no change occurs (that is, $\mathbf{n}_{t+1} = \mathbf{n}_t$). Denoting this value by $\hat{\mathbf{n}}$, it must therefore satisfy the equation

$$\hat{\mathbf{n}} = \begin{bmatrix} 1 & 2 & 4 \\ \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \end{bmatrix} \hat{\mathbf{n}}$$

Find all equilibria and explain what each represents biologically.

- 26. CAT scans** Computed Axial Tomography uses narrow X-ray beams directed through the body at several angles to construct an image of internal organs and structures. The density of the material through which the beam passes affects the extent to which it is diminished before exiting the body. In most cases the part of the body to be scanned is overlaid with a grid, and X-ray beams are directed through the elements of this grid at different angles. Measurements of the exiting beams can then be used to calculate the density of each element of the grid.

(a) Consider a simple grid with three elements, and three X-ray beams directed through them at different angles, as shown in the figure. The unknown densities of the material in each element are x_1 , x_2 and x_3 .



Suppose that, if a beam travels through elements i and j , its intensity upon exiting is $e^{-(x_i+x_j)}$. If the measured intensities of the beams at detectors A, B, and C are a , b , and c , respectively, construct a system of equations for the unknown densities.

(b) Using $a = 0.6$, $b = 0.5$, and $c = 0.7$, determine the density of each element. In CAT scans these densities are then used to color the pixels of the image.

27–30 Find the eigenvectors and eigenvalues for the matrix.

27. $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$

28. $A = \begin{bmatrix} 2 & -2 \\ -4 & 4 \end{bmatrix}$

29. $A = \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$

30. $A = \begin{bmatrix} a & -1 \\ 1 & a \end{bmatrix}$

- 31.** Suppose A is a 2×2 matrix whose columns sum to one. Show that $\lambda = 1$ is an eigenvalue of A .

32–35 Express the solution to the recursion $\mathbf{n}_{t+1} = A\mathbf{n}_t$ in terms of the eigenvectors and eigenvalues of A . Use De Moivre's Theorem to simplify the solution, if appropriate.

32. $A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$ and $\mathbf{n}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

33. $A = \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix}$ and $\mathbf{n}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

34. $A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$ and $\mathbf{n}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

35. $A = \begin{bmatrix} a & -1 \\ 1 & a \end{bmatrix}$ and $\mathbf{n}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, where a is a real number

- 36. Dispersal** In Exercise 8.5.9 we modeled the dispersal of dragonflies using the recursion $\mathbf{x}_{t+1} = A\mathbf{x}_t$, where

$$A = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$$

- Verify that the Perron-Frobenius Theorem can be applied to this model.
- Calculate the eigenvalues of A .
- Given your answers to parts (a) and (b), what is the long-term behavior of $\mathbf{x}_t$?
- Using the initial condition $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, express the solution to the recursion in terms of the eigenvectors and eigenvalues of A .
- Verify your answer to part (c) using the solution you obtained in part (d).

- 37. Succession** In Exercise 8.5.15 we modeled ecological succession using the recursion $\mathbf{y}_{t+1} = A\mathbf{y}_t$, where

$$A = \begin{bmatrix} 0.9 & 0.001 & 0 \\ 0.1 & 0.799 & 0.001 \\ 0 & 0.2 & 0.999 \end{bmatrix}$$

Can the Perron-Frobenius Theorem be applied to this model?

- 38. Inbreeding** In Exercise 8.5.19 we modeled inbreeding using the recursion $\mathbf{x}_{t+1} = A\mathbf{x}_t$, where

$$A = \begin{bmatrix} 1 & \frac{1}{4} & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{4} & 1 \end{bmatrix}$$

- (a) Calculate the eigenvalues of A . (You will find a repeated eigenvalue.)
 (b) Verify that the eigenvectors of A are

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

CAS

- (c) Using the initial condition $\mathbf{x}_0 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, express the solu-

tion to the recursion in terms of the eigenvectors and eigenvalues of A .

- 39.** A sequence is given recursively by the equation

$$a_n = \frac{(a_{n-1} + a_{n-2})}{2}$$

Define the two new variables $x_n = a_n$ and $y_n = a_{n-1}$.

- (a) Find a matrix A such that the vector $\mathbf{z}_n = \begin{bmatrix} x_n \\ y_n \end{bmatrix}$ obeys

the recursion $\mathbf{z}_n = A\mathbf{z}_{n-1}$.

- (b) Verify that the Perron-Frobenius Theorem can be applied.
 (c) Find the eigenvalues of A .
 (d) In light of the Perron-Frobenius Theorem, what is the long-term behavior of the vector $\mathbf{z}_n$?
 (e) Evaluate $\lim_{n \rightarrow \infty} a_n$.

Multivariable Calculus

9

The energy needed by a lizard (such as this iguana) to walk or run depends on both its weight and its speed. So the energy is a function of two variables, which we investigate in Exercise 47 in Section 9.2.

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9.1 Functions of Several Variables

9.2 Partial Derivatives

9.3 Tangent Planes and Linear Approximations

PROJECT: The Speedo LZR Racer

9.4 The Chain Rule

9.5 Directional Derivatives and the Gradient Vector

9.6 Maximum and Minimum Values

BIOLOGICAL QUANTITIES OFTEN DEPEND ON two or more variables. In this chapter we extend the basic ideas of differential calculus to functions of several variables.

9.1 Functions of Several Variables

In this section we study functions of two or more variables from four points of view:

- verbally (by a description in words)
- numerically (by a table of values)
- algebraically (by an explicit formula)
- visually (by a graph or level curves)

■ Functions of Two Variables

The temperature T at a point on the surface of the earth at any given time depends on the longitude x and latitude y of the point. We can think of T as being a function of the two variables x and y , or as a function of the pair (x, y) . We indicate this functional dependence by writing $T = f(x, y)$.

The resistance R of blood flowing through an artery depends on the radius r and length L of the artery. In fact, one of Poiseuille's laws states that $R = CL/r^4$, where C is a positive constant determined by the viscosity of the blood. We say that R is a function of r and L , and we write

$$R(r, L) = C \frac{L}{r^4}$$

Definition A **function f of two variables** is a rule that assigns to each ordered pair of real numbers (x, y) in a set D a unique real number denoted by $f(x, y)$. The set D is the **domain** of f and its **range** is the set of values that f takes on, that is, $\{f(x, y) \mid (x, y) \in D\}$.

We often write $z = f(x, y)$ to make explicit the value taken on by f at the general point (x, y) . The variables x and y are **independent variables** and z is the **dependent variable**. [Compare this with the notation $y = f(x)$ for functions of a single variable.]

The domain is a subset of $\mathbb{R}^2$, the xy -plane. We can think of the domain as the set of all possible inputs and the range as the set of all possible outputs. If a function f is given by a formula and no domain is specified, then the domain of f is understood to be the set of all pairs (x, y) for which the given expression is a well-defined real number.

EXAMPLE 1 | Wind-chill index In regions with severe winter weather, the *wind-chill index* is often used to describe the apparent severity of the cold. This index W is a subjective temperature that depends on the actual temperature T and the wind speed v . So W is a function of T and v , and we can write $W = f(T, v)$. Table 1 records values of W compiled by the National Weather Service of the US and the Meteorological Service of Canada.

Table 1 Wind-chill index as a function of air temperature and wind speed

		Wind speed (km/h)										
$T \backslash v$		5	10	15	20	25	30	40	50	60	70	80
	5	4	3	2	1	1	0	-1	-1	-2	-2	-3
Actual temperature (°C)	0	-2	-3	-4	-5	-6	-6	-7	-8	-9	-9	-10
	-5	-7	-9	-11	-12	-12	-13	-14	-15	-16	-16	-17
	-10	-13	-15	-17	-18	-19	-20	-21	-22	-23	-23	-24
	-15	-19	-21	-23	-24	-25	-26	-27	-29	-30	-30	-31
	-20	-24	-27	-29	-30	-32	-33	-34	-35	-36	-37	-38
	-25	-30	-33	-35	-37	-38	-39	-41	-42	-43	-44	-45
	-30	-36	-39	-41	-43	-44	-46	-48	-49	-50	-51	-52
	-35	-41	-45	-48	-49	-51	-52	-54	-56	-57	-58	-60
	-40	-47	-51	-54	-56	-57	-59	-61	-63	-64	-65	-67

The Wind-Chill Index
A new wind-chill index was introduced in November of 2001 and is more accurate than the old index for measuring how cold it feels when it's windy. The new index is based on a model of how fast a human face loses heat. It was developed through clinical trials in which volunteers were exposed to a variety of temperatures and wind speeds in a refrigerated wind tunnel.

For instance, the table shows that if the temperature is -5°C and the wind speed is 50 km/h, then subjectively it would feel as cold as a temperature of about -15°C with no wind. So

$f(-5, 50) = -15^{\circ}\text{C}$ ■

EXAMPLE 2 | The body mass index (BMI) of a person is defined by the equation

$$B(m, h) = \frac{m}{h^2}$$

where m is the person's mass (in kilograms) and h is the height (in meters). So B is a function of the two variables m and h . A rough guideline is that a person is underweight if the BMI is less than 18.5; optimal if the BMI lies between 18.5 and 25; overweight if the BMI lies between 25 and 30; and obese if the BMI exceeds 30. If someone weighs 64 kg and is 168 cm high, what is the BMI?

SOLUTION With $m = 64$ kg and $h = 1.68$ m, the BMI is

$$B(64, 1.68) = \frac{64}{(1.68)^2} \approx 22.7 \text{ kg/m}^2$$

Because $18.5 \leq B(64, 1.68) \leq 25$

this person is considered to have optimal weight. ■

EXAMPLE 3 | For each of the following functions, evaluate $f(3, 2)$ and find and sketch the domain.

(a) $f(x, y) = \frac{\sqrt{x + y + 1}}{x - 1}$ (b) $f(x, y) = x \ln(y^2 - x)$

SOLUTION

(a) $f(3, 2) = \frac{\sqrt{3 + 2 + 1}}{3 - 1} = \frac{\sqrt{6}}{2}$

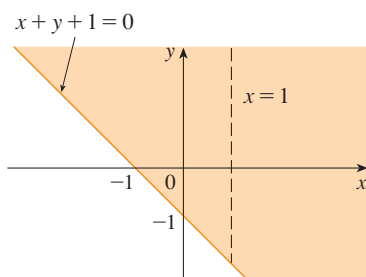


FIGURE 1

Domain of $f(x, y) = \frac{\sqrt{x+y+1}}{x-1}$

The expression for f makes sense if the denominator is not 0 and the quantity under the square root sign is nonnegative. So the domain of f is

$$D = \{(x, y) \mid x + y + 1 \geq 0, x \neq 1\}$$

The inequality $x + y + 1 \geq 0$, or $y \geq -x - 1$, describes the points that lie on or above the line $y = -x - 1$, while $x \neq 1$ means that the points on the line $x = 1$ must be excluded from the domain (see Figure 1).

(b)
$$f(3, 2) = 3 \ln(2^2 - 3) = 3 \ln 1 = 0$$

Since $\ln(y^2 - x)$ is defined only when $y^2 - x > 0$, that is, $x < y^2$, the domain of f is $D = \{(x, y) \mid x < y^2\}$. This is the set of points to the left of the parabola $x = y^2$. (See Figure 2.)

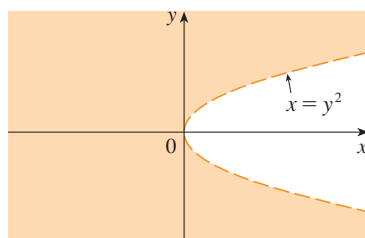


FIGURE 2

Domain of $f(x, y) = x \ln(y^2 - x)$

EXAMPLE 4 | Find the domain and range of $g(x, y) = \sqrt{9 - x^2 - y^2}$.

SOLUTION The domain of g is

$$D = \{(x, y) \mid 9 - x^2 - y^2 \geq 0\} = \{(x, y) \mid x^2 + y^2 \leq 9\}$$

which is the disk with center $(0, 0)$ and radius 3. (See Figure 3.) The range of g is

$$\{z \mid z = \sqrt{9 - x^2 - y^2}, (x, y) \in D\}$$

Since z is a positive square root, $z \geq 0$. Also, because $9 - x^2 - y^2 \leq 9$, we have

$$\sqrt{9 - x^2 - y^2} \leq 3$$

So the range is

$$\{z \mid 0 \leq z \leq 3\} = [0, 3]$$

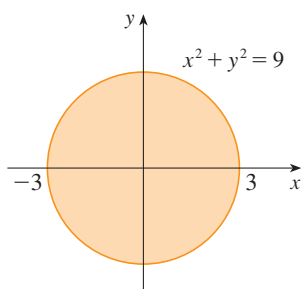


FIGURE 3

Domain of $g(x, y) = \sqrt{9 - x^2 - y^2}$

Graphs

One way of visualizing the behavior of a function of two variables is to consider its graph.

Definition If f is a function of two variables with domain D , then the **graph** of f is the set of all points (x, y, z) in $\mathbb{R}^3$ such that $z = f(x, y)$ and (x, y) is in D .

Just as the graph of a function f of one variable is a curve C with equation $y = f(x)$, so the graph of a function f of two variables is a surface S with equation $z = f(x, y)$. We can visualize the graph S of f as lying directly above or below its domain D in the xy -plane (see Figure 4).

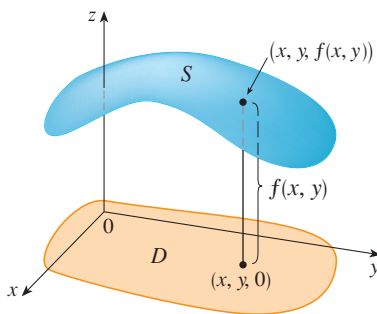


FIGURE 4

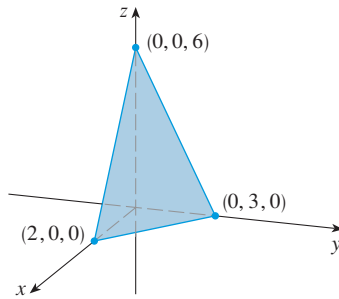


FIGURE 5

EXAMPLE 5 | Sketch the graph of the function $f(x, y) = 6 - 3x - 2y$.

SOLUTION The graph of f has the equation $z = 6 - 3x - 2y$, or $3x + 2y + z = 6$, which represents a plane (see Section 8.3). To graph the plane we first find the intercepts. Putting $y = z = 0$ in the equation, we get $x = 2$ as the x -intercept. Similarly, the y -intercept is 3 and the z -intercept is 6. This helps us sketch the portion of the graph that lies in the first octant in Figure 5.

The function in Example 5 is a special case of the function

$$f(x, y) = ax + by + c$$

which is called a **linear function**. The graph of such a function has the equation

$$z = ax + by + c \quad \text{or} \quad ax + by - z + c = 0$$

so it is a plane. In much the same way that linear functions of one variable are important in single-variable calculus, we will see that linear functions of two variables play a central role in multivariable calculus.

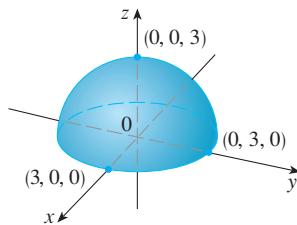


FIGURE 6

EXAMPLE 6 | Sketch the graph of $g(x, y) = \sqrt{9 - x^2 - y^2}$.

SOLUTION The graph has equation $z = \sqrt{9 - x^2 - y^2}$. We square both sides of this equation to obtain $z^2 = 9 - x^2 - y^2$, or $x^2 + y^2 + z^2 = 9$, which we recognize as an equation of the sphere with center the origin and radius 3. (See Section 8.1.) But, since $z \geq 0$, the graph of g is just the top half of this sphere (see Figure 6).

NOTE An entire sphere can't be represented by a single function of x and y . As we saw in Example 6, the upper hemisphere of the sphere $x^2 + y^2 + z^2 = 9$ is represented by the function $g(x, y) = \sqrt{9 - x^2 - y^2}$. The lower hemisphere is represented by the function $h(x, y) = -\sqrt{9 - x^2 - y^2}$.

EXAMPLE 7 | Find the domain and range and sketch the graph of $h(x, y) = x^2 + y^2$.

SOLUTION Notice that $h(x, y)$ is defined for all possible ordered pairs of real numbers (x, y) , so the domain is $\mathbb{R}^2$, the entire xy -plane. The range of h is the set $[0, \infty)$ of all nonnegative real numbers. [Notice that $x^2 \geq 0$ and $y^2 \geq 0$, so $h(x, y) \geq 0$ for all x and y .]

The graph of h has the equation $z = x^2 + y^2$. If we put $x = 0$, we get $z = y^2$, so the yz -plane intersects the surface in a parabola. If we put $x = k$ (a constant), we get $z = y^2 + k^2$. This means that if we slice the graph with any plane parallel to the yz -plane, we obtain a parabola that opens upward. (These curves that we get by slicing a surface with a plane parallel to one of the coordinate planes are called **traces**.) Similarly, if $y = k$, the trace is $z = x^2 + k^2$, which is again a parabola that opens upward. If we put $z = k$, we get the horizontal traces $x^2 + y^2 = k$, which we recognize as a family of circles. Knowing the shapes of the traces, we can sketch the graph of f in Figure 7. Because of the parabolic traces, the surface $z = x^2 + y^2$ is called a **paraboloid**.

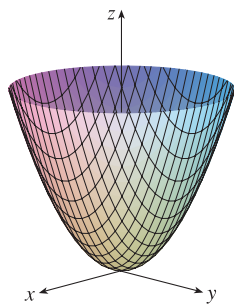


FIGURE 7

Computer programs are readily available for graphing functions of two variables. In most such programs, traces in the vertical planes $x = k$ and $y = k$ are drawn for equally spaced values of k and parts of the graph are eliminated using hidden line removal.

Figure 8 shows computer-generated graphs of several functions. Notice that we get an especially good picture of a function when rotation is used to give views from different vantage points. In parts (a) and (b) the graph of f is very flat and close to the xy -plane except near the origin; this is because $e^{-x^2-y^2}$ is very small when x or y is large.

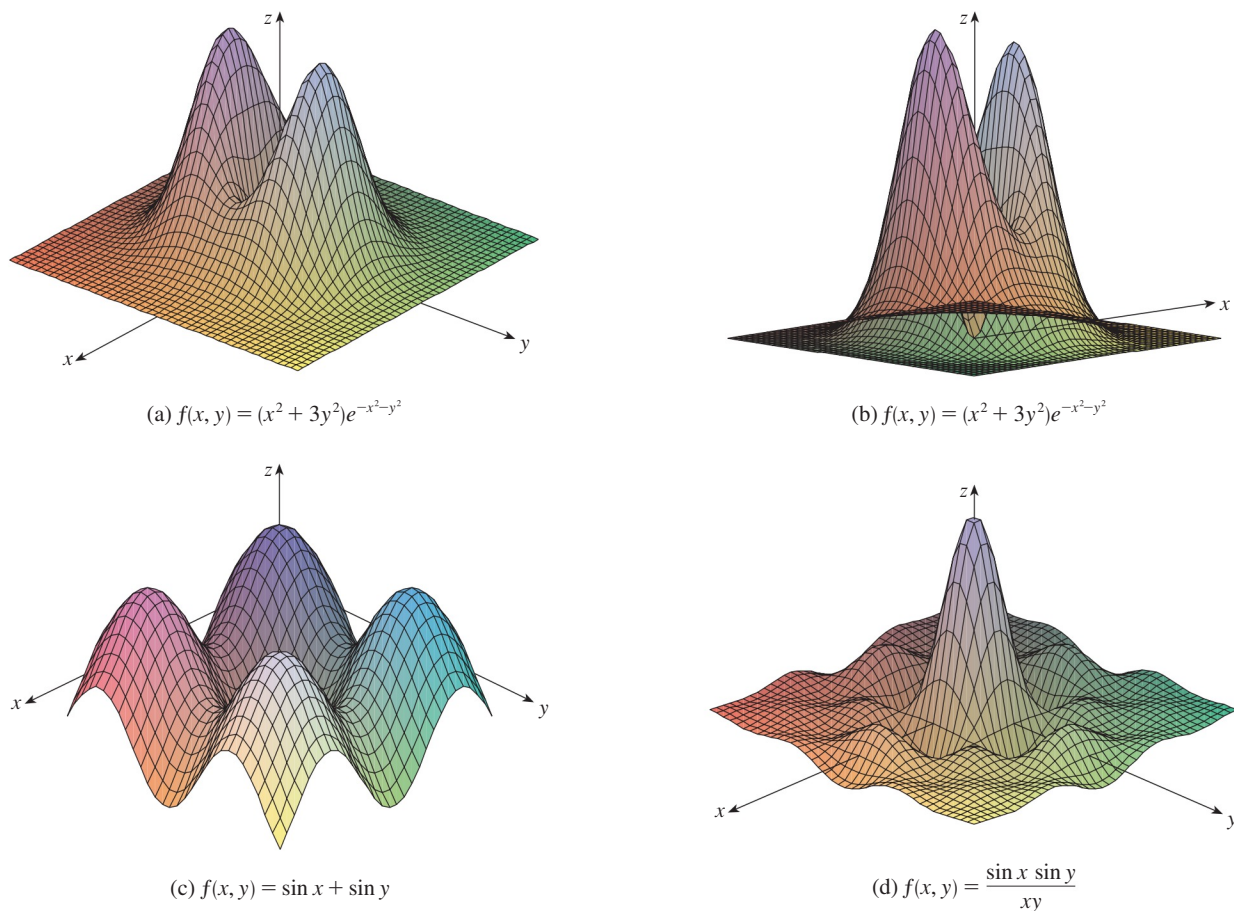


FIGURE 8

■ Level Curves

Another method for visualizing functions, borrowed from mapmakers, is a contour map on which points of constant elevation are joined to form *contour lines*, or *level curves*.

Definition The **level curves** of a function f of two variables are the curves with equations $f(x, y) = k$, where k is a constant (in the range of f).

A level curve $f(x, y) = k$ is the set of all points in the domain of f at which f takes on a given value k . In other words, it shows where the graph of f has height k .

You can see from Figure 9 the relation between level curves and horizontal traces. The level curves $f(x, y) = k$ are just the traces of the graph of f in the horizontal plane $z = k$ projected down to the xy -plane. So if you draw the level curves of a function and visualize them being lifted up to the surface at the indicated height, then you can mentally piece together a picture of the graph. The surface is steep where the level curves are close together. It is somewhat flatter where they are farther apart.

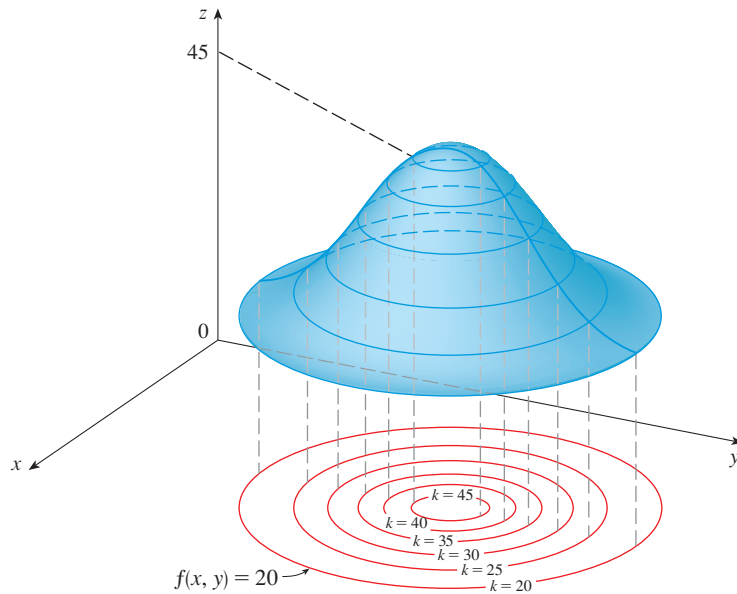


FIGURE 9

TEC Visual 9.1A animates Figure 9 by showing level curves being lifted up to graphs of functions.

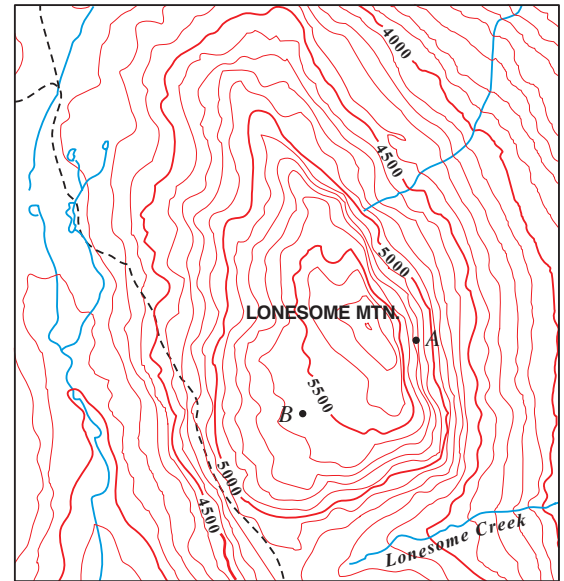
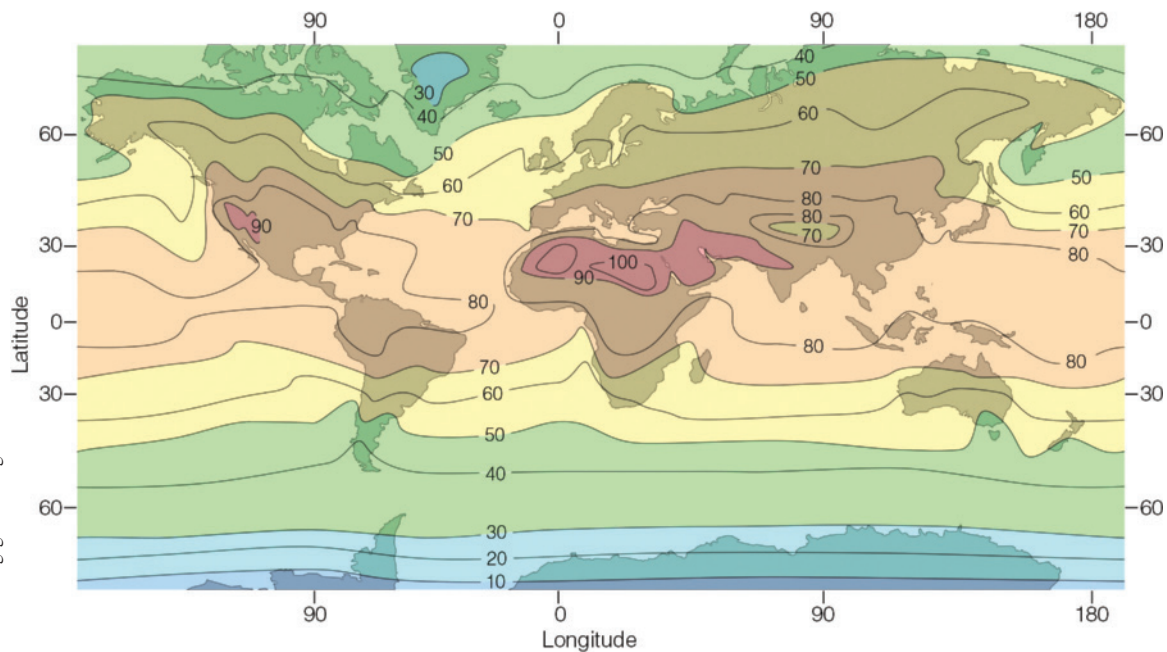


FIGURE 10

One common example of level curves occurs in topographic maps of mountainous regions, such as the map in Figure 10. The level curves are curves of constant elevation above sea level. If you walk along one of these contour lines, you neither ascend nor descend. Another common example is the temperature at locations (x, y) with longitude x and latitude y . Here the level curves are called **isotherms** and join locations with the same temperature. Figure 11 shows a weather map of the world indicating the average July temperatures. The isotherms are the curves that separate the colored bands.

FIGURE 11 Average air temperature near sea level in July ($^{\circ}\text{F}$)

In weather maps of atmospheric pressure at a given time as a function of longitude and latitude, the level curves are called **isobars** and join locations with the same pressure (see Exercise 28). Surface winds tend to flow from areas of high pressure across the isobars toward areas of low pressure, and are strongest where the isobars are tightly packed.

A contour map of worldwide precipitation is shown in Figure 12. Here the level curves are not labeled but they separate the colored regions and the amount of precipitation in each region is indicated in the color key.

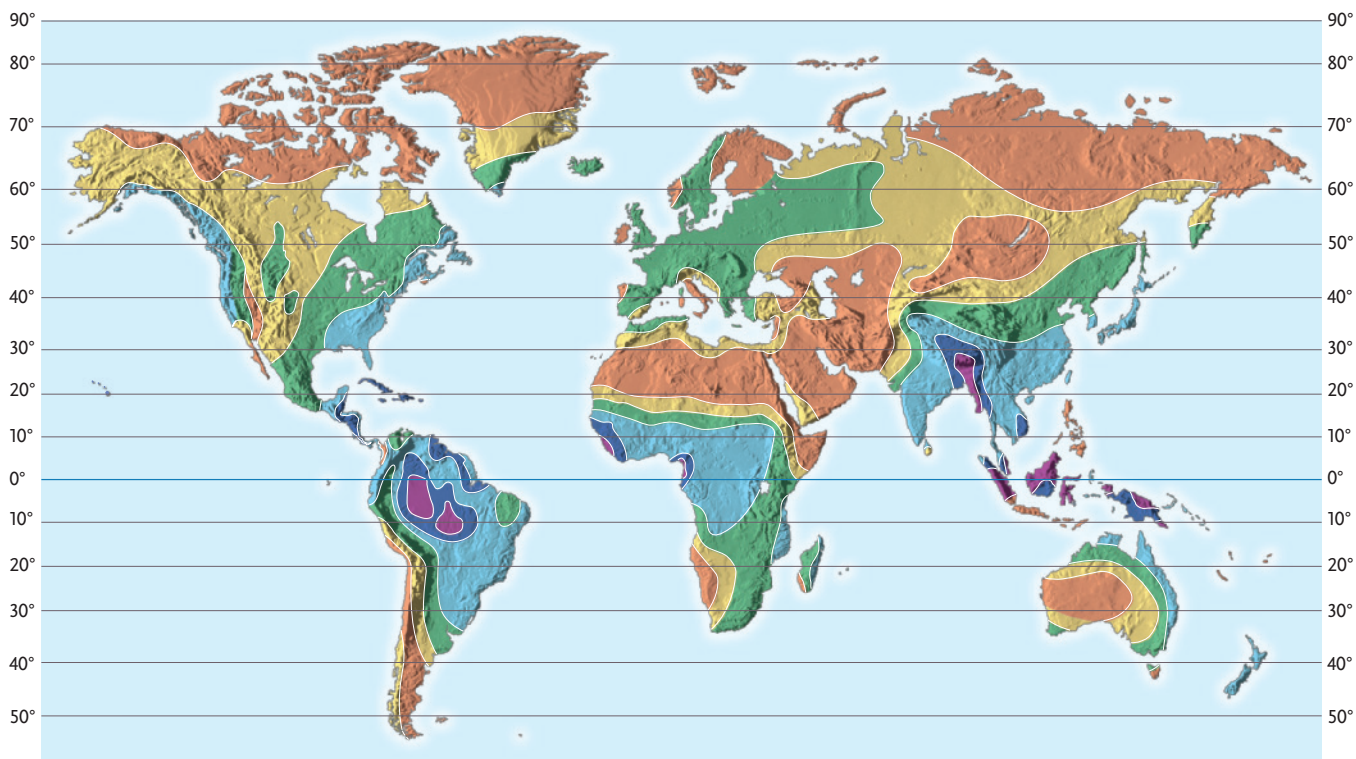


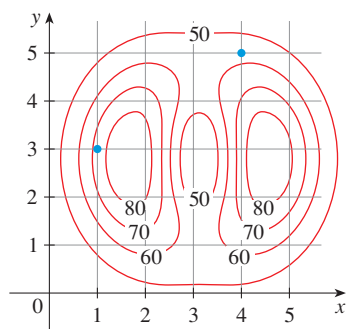
FIGURE 12 Precipitation

From Russell/Hertz/McMillan, *Biology*, 2E. © 2012 Cengage Learning.

KEY

Precipitation (cm/yr)

Under 25	50 to 100	200 to 250
25 to 50	100 to 200	Over 250



EXAMPLE 8 | A contour map for a function f is shown in Figure 13. Use it to estimate the values of $f(1, 3)$ and $f(4, 5)$.

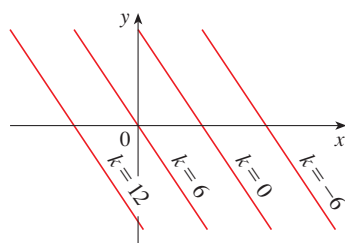
SOLUTION The point $(1, 3)$ lies partway between the level curves with z -values 70 and 80. We estimate that

$$f(1, 3) \approx 73$$

Similarly, we estimate that

$$f(4, 5) \approx 56$$

FIGURE 13

**FIGURE 14**

Contour map of
 $f(x, y) = 6 - 3x - 2y$

EXAMPLE 9 | Sketch the level curves of the function $f(x, y) = 6 - 3x - 2y$ for the values $k = -6, 0, 6, 12$.

SOLUTION The level curves are

$$6 - 3x - 2y = k \quad \text{or} \quad 3x + 2y + (k - 6) = 0$$

This is a family of lines with slope $-\frac{3}{2}$. The four particular level curves with $k = -6, 0, 6$, and 12 are $3x + 2y - 12 = 0$, $3x + 2y - 6 = 0$, $3x + 2y = 0$, and $3x + 2y + 6 = 0$. They are sketched in Figure 14. The level curves are equally spaced parallel lines because the graph of f is a plane (see Figure 5).

EXAMPLE 10 | Sketch the level curves of the function

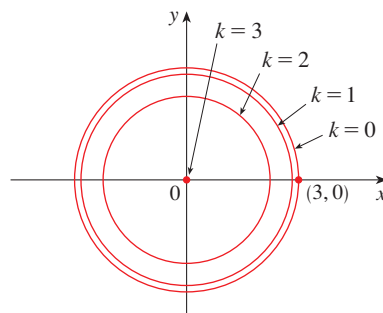
$$g(x, y) = \sqrt{9 - x^2 - y^2} \quad \text{for } k = 0, 1, 2, 3$$

SOLUTION The level curves are

$$\sqrt{9 - x^2 - y^2} = k \quad \text{or} \quad x^2 + y^2 = 9 - k^2$$

This is a family of concentric circles with center $(0, 0)$ and radius $\sqrt{9 - k^2}$. The cases $k = 0, 1, 2, 3$ are shown in Figure 15. Try to visualize these level curves lifted up to form a surface and compare with the graph of g (a hemisphere) in Figure 6. (See TEC Visual 9.1A.)

TEC Visual 9.1B demonstrates the connection between surfaces and their contour maps.

**FIGURE 15**

Contour map of $g(x, y) = \sqrt{9 - x^2 - y^2}$

EXAMPLE 11 | Body mass index (continued) In Example 2 we discussed the body mass index function

$$B(m, h) = \frac{m}{h^2}$$

Sketch some level curves for this function.

SOLUTION The level curves have the equations

$$\frac{m}{h^2} = k \quad \text{or} \quad m = kh^2$$

for various values of k . These are parabolas in the hm -plane but, since $h > 0$ and

$m > 0$, the level curves are the portions that lie in the first quadrant. In Figure 16 these curves are labeled with the value of the BMI. For instance, the curve labeled 20 shows all values of the height h and mass m that result in a BMI of 20.

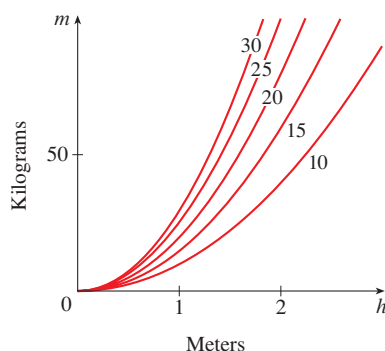


FIGURE 16

Level curves for body mass index

EXAMPLE 12 | **BB** Infectious disease control A quantity of central importance in the spread of infectious diseases is the basic reproduction number R_0 (see Example 5 on page 56). This gives the average number of new infections that each infected individual produces when introduced into a completely susceptible population. Models for the spread of SARS have been constructed to determine the effect of vaccination and quarantine on R_0 .¹ In the simplest case, we have

$$R_0(d, v) = 5(1 - v) \frac{d}{1 + d}$$

where v is the fraction of the population that is vaccinated and d is the average number of days that individuals remain in the population while infectious. (Quarantine reduces this number.)

Figure 17(a) shows a graph of $R_0(d, v)$ drawn by a computer algebra system. Notice how the reproduction number approaches 0 as v approaches 1 (the entire population is vaccinated). From the rotated graph in Figure 17(b) we see how R_0 approaches 5 as d increases and v approaches 0. Thus, qualitatively, the spread of SARS in the population is reduced when the vaccination fraction is high ($v \rightarrow 1$) and when quarantine levels are high ($d \rightarrow 0$). In the coming sections we will explore the effects of these two interventions more quantitatively. The level curves of R_0 are shown in Figure 17(c).

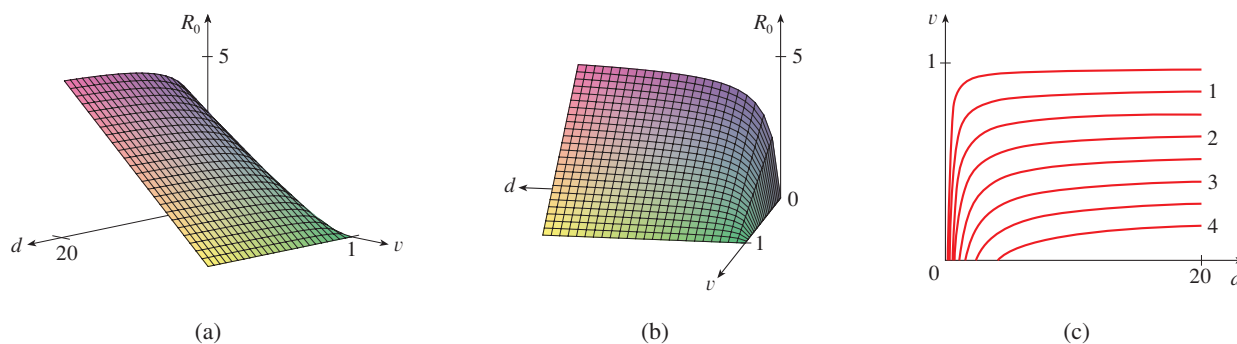
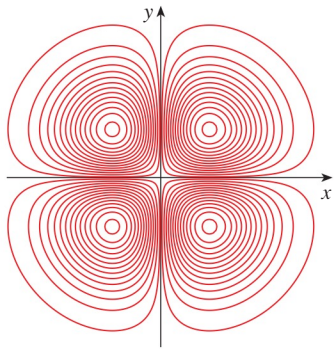


FIGURE 17

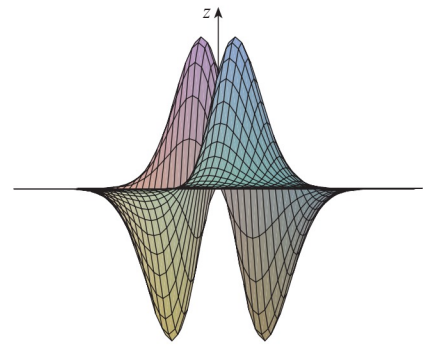
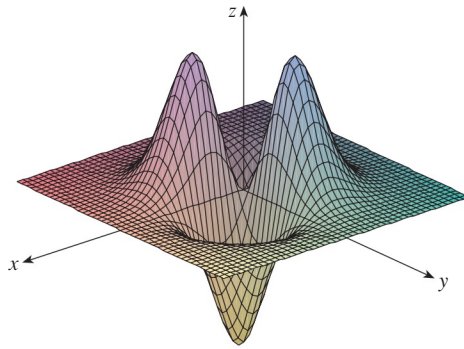
Visualizing the basic reproduction number

1. A. Gumel et al., “Modelling Strategies for Controlling SARS Outbreaks,” *Proceedings of the Royal Society: Series B* 271 (2004): 2223–32.

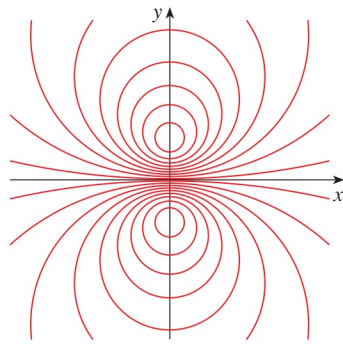
Figure 18 shows some computer-generated level curves together with the corresponding computer-generated graphs. Notice that the level curves in part (c) crowd together near the origin. That corresponds to the fact that the graph in part (d) is very steep near the origin.



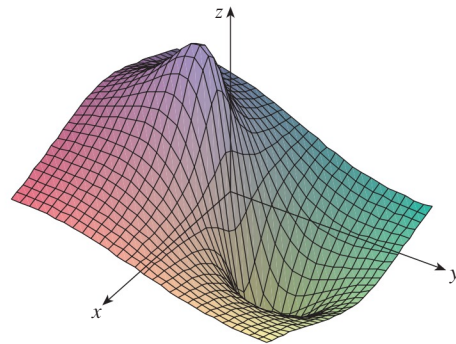
(a) Level curves of $f(x, y) = -xye^{-x^2-y^2}$



(b) Two views of $f(x, y) = -xye^{-x^2-y^2}$



(c) Level curves of $f(x, y) = \frac{-3y}{x^2 + y^2 + 1}$



(d) $f(x, y) = \frac{-3y}{x^2 + y^2 + 1}$

FIGURE 18

■ Functions of Three Variables

A **function of three variables**, f , is a rule that assigns to each ordered triple (x, y, z) in a domain $D \subset \mathbb{R}^3$ a unique real number denoted by $f(x, y, z)$. For instance, the temperature T at a point on the surface of the earth depends on the longitude x and latitude y of the point and on the time t , so we could write $T = f(x, y, t)$.

EXAMPLE 13 | When migrating fish swim at a speed v relative to the water, the energy expenditure per unit time is proportional to v^3 . Suppose the fish swim for a distance d against a current that has speed u , where $u < v$. Write the required energy E as a function of the three variables u , v , and d .

SOLUTION When swimming against the current, the ground speed of the fish is $v - u$. If T is the time required to travel the ground distance d , then $d = (v - u)T$, so

$$T = \frac{d}{v - u}$$

The energy per unit time is av^3 , where a is the proportionality constant. So the total energy is

$$E = av^3 \cdot T = av^3 \cdot \frac{d}{v - u}$$

Thus the energy as a function of u , v , and d is

$$E(u, v, d) = \frac{av^3d}{v - u}$$

EXAMPLE 14 | Find the domain of f if

$$f(x, y, z) = \ln(z - y) + xy \sin z$$

SOLUTION The expression for $f(x, y, z)$ is defined as long as $z - y > 0$, so the domain of f is

$$D = \{(x, y, z) \in \mathbb{R}^3 \mid z > y\}$$

This is a **half-space** consisting of all points that lie above the plane $z = y$.

Functions of any number of variables can be considered. A **function of n variables** is a rule that assigns a number $z = f(x_1, x_2, \dots, x_n)$ to an n -tuple $(x_1, x_2, \dots, x_n)$ of real numbers. We denote by $\mathbb{R}^n$ the set of all such n -tuples. For example, if a company uses n different ingredients in making a food product, c_i is the cost per unit of the i th ingredient, and x_i units of the i th ingredient are used, then the total cost C of the ingredients is a function of the n variables $x_1, x_2, \dots, x_n$:

$$C = f(x_1, x_2, \dots, x_n) = c_1x_1 + c_2x_2 + \cdots + c_nx_n$$

■ Limits and Continuity

For functions of a single variable, recall from Section 2.3 the meaning of a limit: $\lim_{x \rightarrow a} f(x) = L$ means that we can make the values of $f(x)$ as close to L as we like by taking x to be sufficiently close to a , but not equal to a . For functions of two variables we use the notation

$$\lim_{(x, y) \rightarrow (a, b)} f(x, y) = L$$

to indicate that the values of $f(x, y)$ approach the number L as the point (x, y) approaches the point (a, b) along any path that stays within the domain of f .

(1) Definition We write

$$\lim_{(x, y) \rightarrow (a, b)} f(x, y) = L$$

and we say that the **limit of $f(x, y)$ as (x, y) approaches (a, b)** is L if we can make the values of $f(x, y)$ as close to L as we like by taking the point (x, y) sufficiently close to the point (a, b) , but not equal to (a, b) .

A more precise definition of the limit of a function of two variables is given in Appendix D.

For functions of a single variable, when we let x approach a , there are only two possible directions of approach, from the left or from the right. We recall from Chapter 2 that if $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$, then $\lim_{x \rightarrow a} f(x)$ does not exist.

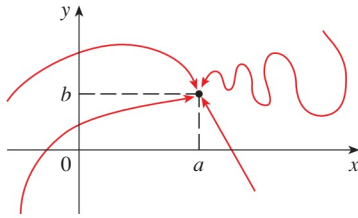


FIGURE 19

For functions of two variables the situation is not as simple because we can let (x, y) approach (a, b) from an infinite number of directions in any manner whatsoever (see Figure 19) as long as (x, y) stays within the domain of f .

Definition 1 says that the distance between $f(x, y)$ and L can be made arbitrarily small by making the distance from (x, y) to (a, b) sufficiently small (but not 0). The definition refers only to the *distance* between (x, y) and (a, b) . It does not refer to the direction of approach. Therefore, if the limit exists, then $f(x, y)$ must approach the same limit no matter how (x, y) approaches (a, b) . Thus, if we can find two different paths of approach along which the function $f(x, y)$ has different limits, then it follows that $\lim_{(x, y) \rightarrow (a, b)} f(x, y)$ does not exist.

If $f(x, y) \rightarrow L_1$ as $(x, y) \rightarrow (a, b)$ along a path C_1 and $f(x, y) \rightarrow L_2$ as $(x, y) \rightarrow (a, b)$ along a path C_2 , where $L_1 \neq L_2$, then $\lim_{(x, y) \rightarrow (a, b)} f(x, y)$ does not exist.

EXAMPLE 15 | If $f(x, y) = xy/(x^2 + y^2)$, does $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ exist?

SOLUTION First let's approach $(0, 0)$ along the x -axis. If $y = 0$, then $f(x, 0) = 0/x^2 = 0$. Therefore

$$f(x, y) \rightarrow 0 \quad \text{as} \quad (x, y) \rightarrow (0, 0) \text{ along the } x\text{-axis}$$

If $x = 0$, then $f(0, y) = 0/y^2 = 0$, so

$$f(x, y) \rightarrow 0 \quad \text{as} \quad (x, y) \rightarrow (0, 0) \text{ along the } y\text{-axis}$$

Although we have obtained identical limits along the axes, that does not show that the given limit is 0. Let's now approach $(0, 0)$ along another line, say $y = x$. For all $x \neq 0$,

$$f(x, x) = \frac{x^2}{x^2 + x^2} = \frac{1}{2}$$

Therefore $f(x, y) \rightarrow \frac{1}{2}$ as $(x, y) \rightarrow (0, 0)$ along $y = x$

(See Figure 20.) Since we have obtained different limits along different paths, the given limit does not exist. ■

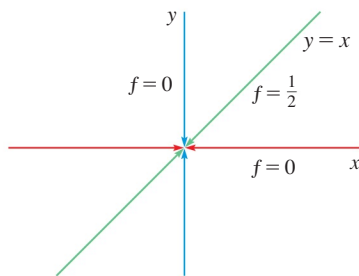
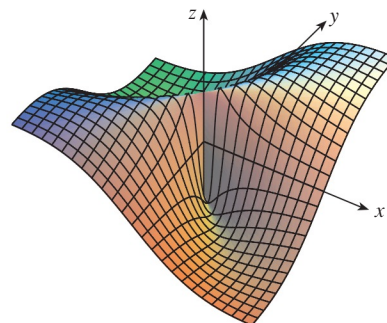


FIGURE 20

TEC In Visual 9.1C a rotating line on the surface in Figure 21 shows different limits at the origin from different directions.

FIGURE 21

$$f(x, y) = \frac{xy}{x^2 + y^2}$$



Just as for functions of one variable, the calculation of limits for functions of two variables can be greatly simplified by the use of properties of limits. The Limit Laws listed in Section 2.4 can be extended to functions of two variables: The limit of a sum is the sum of the limits, the limit of a product is the product of the limits, and so on. In particular, the following equations are true.

$$(2) \quad \lim_{(x,y) \rightarrow (a,b)} x = a \quad \lim_{(x,y) \rightarrow (a,b)} y = b \quad \lim_{(x,y) \rightarrow (a,b)} c = c$$

The Squeeze Theorem also holds.

EXAMPLE 16 | Find $\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^2 + y^2}$ if it exists.

SOLUTION If we let $(x, y) \rightarrow (0, 0)$ along any line through the origin, we find that $f(x, y) \rightarrow 0$. This doesn't prove that the given limit is 0, but let's look at the distance from $f(x, y)$ to 0:

$$\left| \frac{3x^2y}{x^2 + y^2} - 0 \right| = \left| \frac{3x^2y}{x^2 + y^2} \right| = \frac{3x^2|y|}{x^2 + y^2}$$

Notice that $x^2 \leq x^2 + y^2$ because $y^2 \geq 0$. So

$$\frac{x^2}{x^2 + y^2} \leq 1$$

$$\text{Thus} \quad 0 \leq \frac{3x^2|y|}{x^2 + y^2} \leq 3|y|$$

Now we use the Squeeze Theorem. Since

$$\lim_{(x,y) \rightarrow (0,0)} 0 = 0 \quad \text{and} \quad \lim_{(x,y) \rightarrow (0,0)} 3|y| = 0 \quad [\text{by (2)}]$$

$$\text{we conclude that} \quad \lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^2 + y^2} = 0$$

Recall that evaluating limits of *continuous* functions of a single variable is easy. It can be accomplished by direct substitution because the defining property of a continuous function is $\lim_{x \rightarrow a} f(x) = f(a)$. Continuous functions of two variables are also defined by the direct substitution property.

(3) Definition A function f of two variables is called **continuous at** (a, b) if

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b)$$

We say f is **continuous on** D if f is continuous at every point (a, b) in D .

The intuitive meaning of continuity is that if the point (x, y) changes by a small amount, then the value of $f(x, y)$ changes by a small amount. This means that a surface that is the graph of a continuous function has no hole or break.

Using the properties of limits, you can see that sums, differences, products, and quotients of continuous functions are continuous on their domains. Let's use this fact to give examples of continuous functions.

A **polynomial function of two variables** (or polynomial, for short) is a sum of terms of the form $cx^m y^n$, where c is a constant and m and n are nonnegative integers. A **rational function** is a ratio of polynomials. For instance,

$$f(x, y) = x^4 + 5x^3 y^2 + 6xy^4 - 7y + 6$$

is a polynomial, whereas

$$g(x, y) = \frac{2xy + 1}{x^2 + y^2}$$

is a rational function.

The limits in (2) show that the functions $f(x, y) = x$, $g(x, y) = y$, and $h(x, y) = c$ are continuous. Since any polynomial can be built up out of the simple functions f , g , and h by multiplication and addition, it follows that *all polynomials are continuous on $\mathbb{R}^2$* . Likewise, any rational function is continuous on its domain because it is a quotient of continuous functions.

EXAMPLE 17 | Evaluate $\lim_{(x, y) \rightarrow (1, 2)} (x^2 y^3 - x^3 y^2 + 3x + 2y)$.

SOLUTION Since $f(x, y) = x^2 y^3 - x^3 y^2 + 3x + 2y$ is a polynomial, it is continuous everywhere, so we can find the limit by direct substitution:

$$\lim_{(x, y) \rightarrow (1, 2)} (x^2 y^3 - x^3 y^2 + 3x + 2y) = 1^2 \cdot 2^3 - 1^3 \cdot 2^2 + 3 \cdot 1 + 2 \cdot 2 = 11 \quad \blacksquare$$

Figure 22 shows the graph of the continuous function in Example 18.

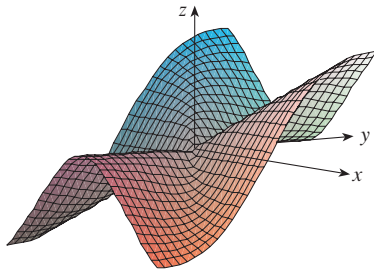


FIGURE 22

EXAMPLE 18 | Let

$$f(x, y) = \begin{cases} \frac{3x^2y}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

We know f is continuous for $(x, y) \neq (0, 0)$ since it is equal to a rational function there. Also, from Example 16, we have

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = \lim_{(x, y) \rightarrow (0, 0)} \frac{3x^2y}{x^2 + y^2} = 0 = f(0, 0)$$

Therefore f is continuous at $(0, 0)$, and so it is continuous on $\mathbb{R}^2$. ■

Just as for functions of one variable, composition is another way of combining two continuous functions to get a third. In fact, it can be shown that if f is a continuous function of two variables and g is a continuous function of a single variable that is defined on the range of f , then the composite function $h = g \circ f$ defined by $h(x, y) = g(f(x, y))$ is also a continuous function.

EXAMPLE 19 | Where is the function $h(x, y) = \arctan(y/x)$ continuous?

SOLUTION The function $f(x, y) = y/x$ is a rational function and therefore continuous except on the line $x = 0$. The function $g(t) = \arctan t$ is continuous everywhere. So the composite function

$$g(f(x, y)) = \arctan(y/x) = h(x, y)$$

is continuous except where $x = 0$. Figure 23 shows the break in the graph of h above the y -axis.

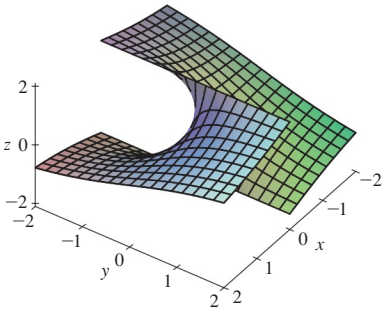


FIGURE 23
The function $h(x, y) = \arctan(y/x)$ is discontinuous where $x = 0$.

Everything that we have done in this section can be extended to functions of three or more variables. The notation

$$\lim_{(x, y, z) \rightarrow (a, b, c)} f(x, y, z) = L$$

means that the values of $f(x, y, z)$ approach the number L as the point (x, y, z) approaches the point (a, b, c) along any path in the domain of f . The function f is **continuous** at (a, b, c) if

$$\lim_{(x, y, z) \rightarrow (a, b, c)} f(x, y, z) = f(a, b, c)$$

For instance, the function

$$f(x, y, z) = \frac{1}{x^2 + y^2 + z^2 - 1}$$

is a rational function of three variables and so is continuous at every point in $\mathbb{R}^3$ except where $x^2 + y^2 + z^2 = 1$. In other words, it is discontinuous on the sphere with center the origin and radius 1.

EXERCISES 9.1

1. **Wind chill** In Example 1 we considered the function $W = f(T, v)$, where W is the wind-chill index, T is the actual temperature, and v is the wind speed. A numerical representation is given in Table 1.
- (a) What is the value of $f(-15, 40)$? What is its meaning?
 - (b) Describe in words the meaning of the question “For what value of v is $f(-20, v) = -30$?” Then answer the question.
 - (c) Describe in words the meaning of the question “For what value of T is $f(T, 20) = -49$?” Then answer the question.
 - (d) What is the meaning of the function $W = f(-5, v)$? Describe the behavior of this function.
 - (e) What is the meaning of the function $W = f(T, 50)$? Describe the behavior of this function.
2. The **temperature-humidity index** I (or humidex, for short) is the perceived air temperature when the actual temperature is T and the relative humidity is h , so we can write $I = f(T, h)$. The following table of values of I is an excerpt from a table compiled by the National Oceanic & Atmospheric Administration.

Table 2 Apparent temperature as a function of temperature and humidity

		Relative humidity (%)					
Actual temperature (°F)	h	20	30	40	50	60	70
	80	77	78	79	81	82	83
	85	82	84	86	88	90	93
	90	87	90	93	96	100	106
	95	93	96	101	107	114	124
	100	99	104	110	120	132	144

- (a) What is the value of $f(95, 70)$? What is its meaning?
- (b) For what value of h is $f(90, h) = 100$?
- (c) For what value of T is $f(T, 50) = 88$?
- (d) What are the meanings of the functions $I = f(80, h)$ and $I = f(100, h)$? Compare the behavior of these two functions of h .

- 3. Body surface area** A model for the surface area of a human body is given by the function

$$S = f(w, h) = 0.1091w^{0.425}h^{0.725}$$

where w is the weight (in pounds), h is the height (in inches), and S is measured in square feet.

- (a) Find $f(160, 70)$ and interpret it.
 (b) What is your own surface area?
- 4. The wind-chill index** W discussed in Example 1 has been modeled by the following function:

$$W(T, v) = 13.12 + 0.6215T - 11.37v^{0.16} + 0.3965Tv^{0.16}$$

Check to see how closely this model agrees with the values in Table 1 for a few values of T and v .

- 5. A manufacturer has modeled its yearly production function** P (the monetary value of its entire production) as a so-called Cobb-Douglas function

$$P(L, K) = 1.47L^{0.65}K^{0.35}$$

where L is the number of labor hours (in thousands) and K is the invested capital (in millions of dollars).

- (a) Find $P(120, 20)$ and interpret it.
 (b) If both the amount of labor and the amount of capital are doubled, verify that the production is also doubled.
- 6. A company makes two kinds of chocolate bars: plain, and with almonds.** Fixed production costs are \$10,000 and it costs \$1.10 to make a plain chocolate bar and \$1.25 to make one with almonds.
- (a) Express the cost of making x plain bars and y bars with almonds as a function of two variables $C = f(x, y)$.
 (b) Find $f(2000, 1000)$ and interpret it.
 (c) What is the domain of f ?

- 7. Snake reversals and stripes** In a study of the survivorship of juvenile garter snakes, a researcher arrived at the model

$$F = 4.2 + 0.008R + 0.102S + 0.017R^2 - 0.034S^2 - 0.268RS$$

where F is a measure of the fitness of the snake, R measures the number of reversals of direction during flight from a predator, and S measures the degree of stripedness in the color pattern of the snake. Which is likelier to survive longer, a snake with $R = 3$ and $S = 1$ or one with $R = 1$ and $S = 3$?

Source: Adapted from E. Brodie III, "Correlational Selection for Color Pattern and Antipredator Behavior in the Garter Snake *Thamnophis Ordinoideis*," *Evolution* 46 (1992): 1284–98.



- 8. Blood flow** The shape of a blood vessel (a vein or artery) can be modeled by a cylindrical tube with radius R and length L . The velocity v of the blood is modeled by Poiseuille's law of laminar flow, which expresses v as a function of five variables:

$$v = f(P, \eta, L, R, r) = \frac{P}{4\eta L} (R^2 - r^2)$$

where η is the viscosity of the blood, P is the pressure difference between the ends of the tube (in dynes/cm²), r is the distance from the central axis of the tube, and r , R , and L are measured in centimeters.

- (a) Evaluate $f(4000, 0.027, 2, 0.008, 0.002)$ and interpret it. (These values are typical for some of the smaller human arteries.)
 (b) Where in the artery is the flow the greatest? Where is it least?
- 9. Let** $g(x, y) = \cos(x + 2y)$.
 (a) Evaluate $g(2, -1)$.
 (b) Find the domain of g .
 (c) Find the range of g .
- 10. Let** $F(x, y) = 1 + \sqrt{4 - y^2}$.
 (a) Evaluate $F(3, 1)$.
 (b) Find and sketch the domain of F .
 (c) Find the range of F .
- 11. Let** $f(x, y, z) = \sqrt{x} + \sqrt{y} + \sqrt{z} + \ln(4 - x^2 - y^2 - z^2)$.
 (a) Evaluate $f(1, 1, 1)$.
 (b) Find and describe the domain of f .
- 12. Let** $g(x, y, z) = x^3y^2z\sqrt{10 - x - y - z}$.
 (a) Evaluate $g(1, 2, 3)$.
 (b) Find and describe the domain of g .

13–20 Find and sketch the domain of the function.

13. $f(x, y) = \sqrt{2x - y}$ **14.** $f(x, y) = \sqrt{xy}$

15. $f(x, y) = \sqrt{1 - x^2} - \sqrt{1 - y^2}$

16. $f(x, y) = \ln(x^2 + y^2 - 2)$

17. $f(x, y) = \frac{\sqrt{y - x^2}}{1 - x^2}$

18. $f(x, y) = \sqrt{y} + \sqrt{25 - x^2 - y^2}$

19. $f(x, y, z) = \sqrt{1 - x^2 - y^2 - z^2}$

20. $f(x, y, z) = \sqrt{z} + \ln(1 - x^2 - y^2)$

21–25 Sketch the graph of the function.

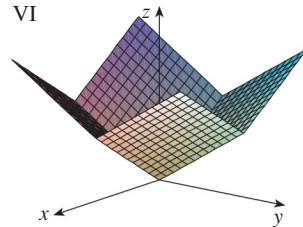
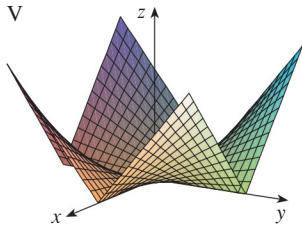
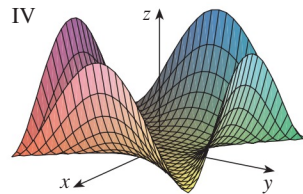
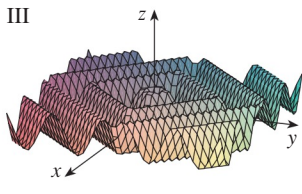
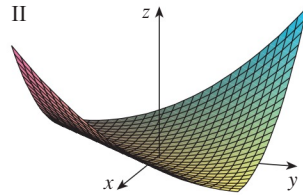
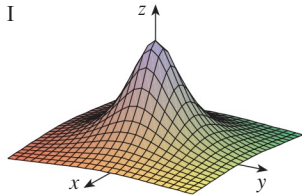
21. $f(x, y) = 3$ **22.** $f(x, y) = y$

23. $f(x, y) = 10 - 4x - 5y$ **24.** $f(x, y) = 1 + 2x^2 + 2y^2$

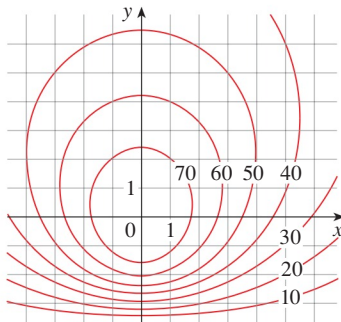
25. $f(x, y) = y^2 + 1$

26. Match the function with its graph (labeled I–VI). Give reasons for your choices.

(a) $f(x, y) = |x| + |y|$ (b) $f(x, y) = |xy|$
 (c) $f(x, y) = \frac{1}{1 + x^2 + y^2}$ (d) $f(x, y) = (x^2 - y^2)^2$
 (e) $f(x, y) = (x - y)^2$ (f) $f(x, y) = \sin(|x| + |y|)$

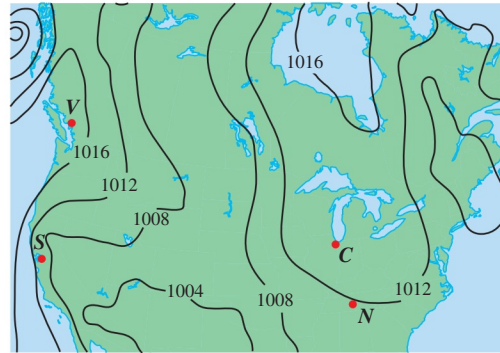


27. A contour map for a function f is shown. Use it to estimate the values of $f(-3, 3)$ and $f(3, -2)$. What can you say about the shape of the graph?

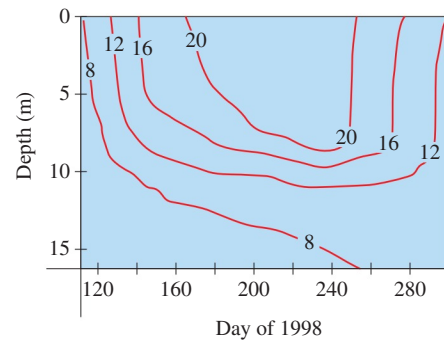


28. Shown is a contour map of atmospheric pressure in North America on August 12, 2008. On the level curves (called isobars) the pressure is indicated in millibars (mb).
 (a) Estimate the pressure at C (Chicago), N (Nashville), S (San Francisco), and V (Vancouver).

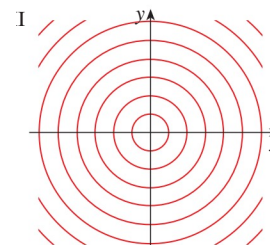
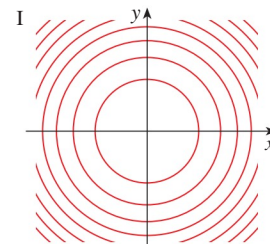
- (b) At which of these locations were the winds strongest?



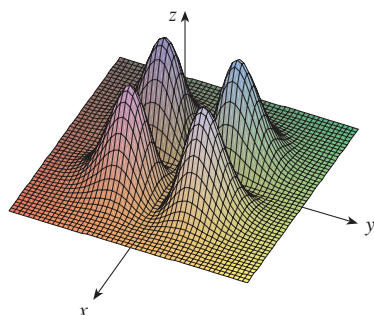
29. Level curves (isotherms) are shown for the water temperature (in $^{\circ}\text{C}$) in Long Lake (Minnesota) in 1998 as a function of depth and time of year. Estimate the temperature in the lake on June 9 (day 160) at a depth of 10 m and on June 29 (day 180) at a depth of 5 m.



30. Two contour maps are shown. One is for a function f whose graph is a cone. The other is for a function g whose graph is a paraboloid. Which is which, and why?



31. Locate the points A and B on the map of Lonesome Mountain (see Figure 10). How would you describe the terrain near A ? Near B ?
32. Make a rough sketch of a contour map for the function whose graph is shown.



33–38 Draw a contour map of the function showing several level curves.

33. $f(x, y) = 2x - y$ 34. $f(x, y) = y - x^2$
35. $f(x, y) = xy$
36. $f(x, y) = \sqrt{36 - x^2 - y^2}$
37. $f(x, y) = ye^x$ 38. $f(x, y) = \frac{y}{x + 1}$

39. Body mass index was discussed in Examples 2 and 11. Draw the level curves $B(m, h) = 18.5$, $B(m, h) = 25$, $B(m, h) = 30$, and $B(m, h) = 40$. Then shade the region corresponding to optimal BMI. Does someone who weighs 62 kg and is 152 cm tall fall into this category?

40. Body mass index Draw the level curve of the body mass index function corresponding to someone who is 200 cm tall and weighs 80 kg. Find the weights and heights of two other people with that same level curve.

41–46 Match the function (a) with its graph (labeled A–F on page 584) and (b) with its contour map (labeled I–VI). Give reasons for your choices.

41. $z = \sin(xy)$ 42. $z = e^x \cos y$
43. $z = \sin(x - y)$ 44. $z = \sin x - \sin y$
45. $z = (1 - x^2)(1 - y^2)$
46. $z = \frac{x - y}{1 + x^2 + y^2}$

CAS 47. Snake reversals and stripes Use computer software to graph the fitness function F in Exercise 7 as well as a contour map. How would you describe the shape of the surface?

CAS 48. Use computer software to graph the function

$$f(x, y) = xy^2 - x^3$$

Why do you think the surface is called a monkey saddle?

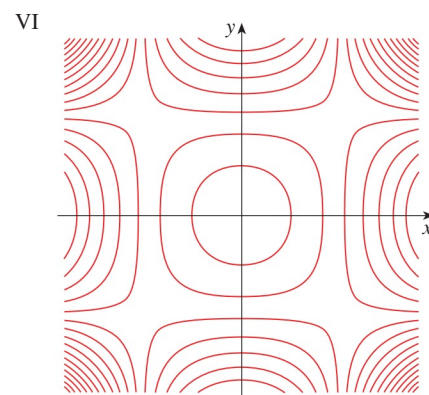
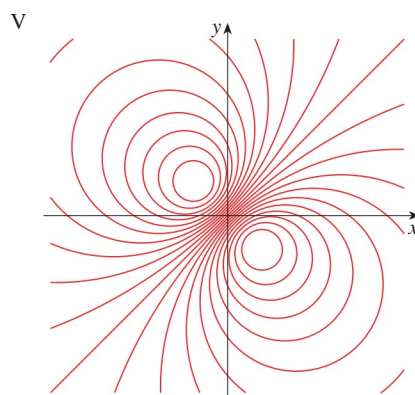
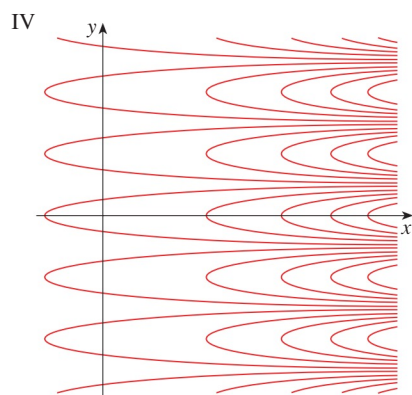
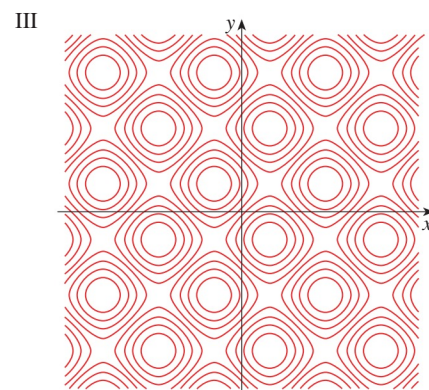
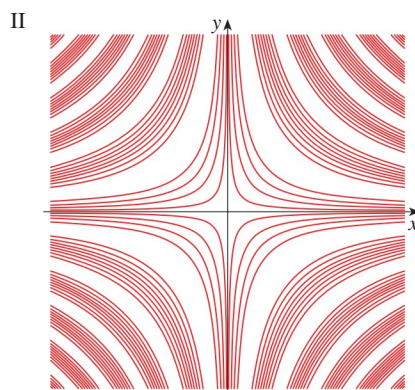
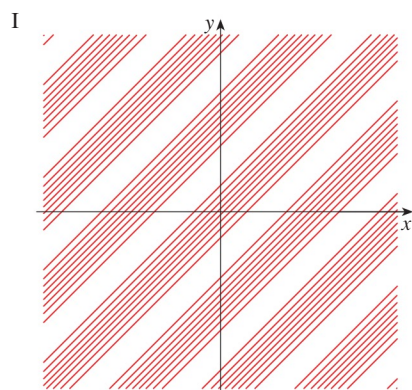
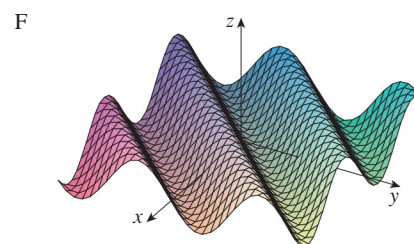
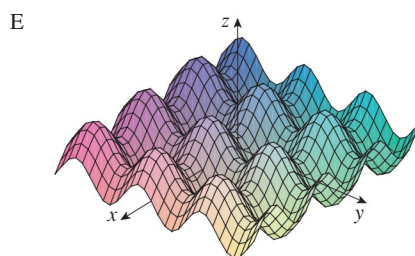
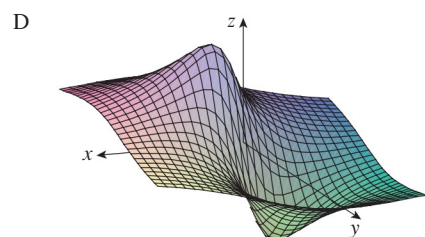
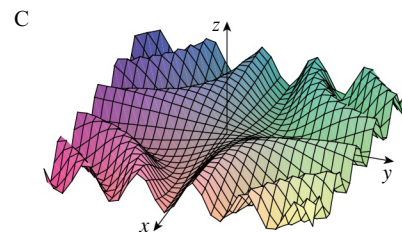
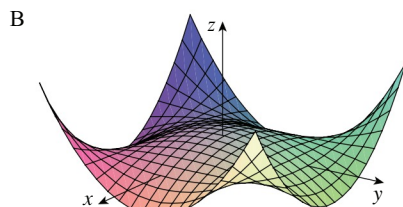
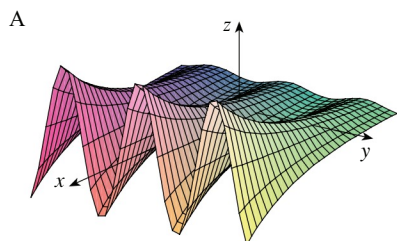
49–58 Find the limit, if it exists, or show that the limit does not exist.

49. $\lim_{(x, y) \rightarrow (1, 2)} (5x^3 - x^2y^2)$
50. $\lim_{(x, y) \rightarrow (1, -1)} e^{-xy} \cos(x + y)$
51. $\lim_{(x, y) \rightarrow (2, 1)} \frac{4 - xy}{x^2 + 3y^2}$ 52. $\lim_{(x, y, z) \rightarrow (\pi, 0, 1/3)} e^{y^2} \tan(xz)$
53. $\lim_{(x, y) \rightarrow (0, 0)} \frac{y^4}{x^4 + 3y^4}$ 54. $\lim_{(x, y) \rightarrow (0, 0)} \frac{x^2 + \sin^2 y}{2x^2 + y^2}$
55. $\lim_{(x, y) \rightarrow (0, 0)} \frac{xy \cos y}{3x^2 + y^2}$ 56. $\lim_{(x, y) \rightarrow (0, 0)} \frac{6x^3y}{2x^4 + y^4}$
57. $\lim_{(x, y) \rightarrow (0, 0)} \frac{xy}{\sqrt{x^2 + y^2}}$ 58. $\lim_{(x, y) \rightarrow (0, 0)} \frac{x^2 \sin^2 y}{x^2 + 2y^2}$

59–66 Determine the set of points at which the function is continuous.

59. $F(x, y) = \arctan(x + \sqrt{y})$
60. $F(x, y) = \cos \sqrt{1 + x - y}$
61. $G(x, y) = \ln(x^2 + y^2 - 4)$
62. $H(x, y) = \frac{e^x + e^y}{e^{xy} - 1}$
63. $f(x, y, z) = \frac{\sqrt{y}}{x^2 - y^2 + z^2}$
64. $f(x, y, z) = \sqrt{x + y + z}$
65. $f(x, y) = \begin{cases} \frac{x^2y^3}{2x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 1 & \text{if } (x, y) = (0, 0) \end{cases}$
66. $f(x, y) = \begin{cases} \frac{xy}{x^2 + xy + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$

Graphs and Contour Maps for Exercises 41–46



9.2 Partial Derivatives

On a hot day, extreme humidity makes us think the temperature is higher than it really is, whereas in very dry air we perceive the temperature to be lower than the thermometer indicates. The National Weather Service has devised the *heat index* (also called the temperature-humidity index, or humidex, in some countries) to describe the combined effects of temperature and humidity. The heat index I is the perceived air temperature when the actual temperature is T and the relative humidity is H . So I is a function of T and H and we can write $I = f(T, H)$. The following table of values of I is an excerpt from a table compiled by the National Weather Service.

Table 1 Heat index I as a function of temperature and humidity

		Relative humidity (%)								
Actual temperature (°F)	H	50	55	60	65	70	75	80	85	90
	90	96	98	100	103	106	109	112	115	119
	92	100	103	105	108	112	115	119	123	128
	94	104	107	111	114	118	122	127	132	137
	96	109	113	116	121	125	130	135	141	146
	98	114	118	123	127	133	138	144	150	157
	100	119	124	129	135	141	147	154	161	168

If we concentrate on the highlighted column of the table, which corresponds to a relative humidity of $H = 70\%$, we are considering the heat index as a function of the single variable T for a fixed value of H . Let's write $g(T) = f(T, 70)$. Then $g(T)$ describes how the heat index I increases as the actual temperature T increases when the relative humidity is 70%. The derivative of g when $T = 96^\circ\text{F}$ is the rate of change of I with respect to T when $T = 96^\circ\text{F}$:

$$g'(96) = \lim_{h \rightarrow 0} \frac{g(96 + h) - g(96)}{h} = \lim_{h \rightarrow 0} \frac{f(96 + h, 70) - f(96, 70)}{h}$$

We can approximate $g'(96)$ using the values in Table 1 by taking $h = 2$ and -2 :

$$\begin{aligned} g'(96) &\approx \frac{g(98) - g(96)}{2} = \frac{f(98, 70) - f(96, 70)}{2} = \frac{133 - 125}{2} = 4 \\ g'(96) &\approx \frac{g(94) - g(96)}{-2} = \frac{f(94, 70) - f(96, 70)}{-2} = \frac{118 - 125}{-2} = 3.5 \end{aligned}$$

Averaging these values, we can say that the derivative $g'(96)$ is approximately 3.75. This means that, when the actual temperature is 96°F and the relative humidity is 70%, the apparent temperature (heat index) rises by about 3.75°F for every degree that the actual temperature rises!

Now let's look at the highlighted row in Table 1, which corresponds to a fixed temperature of $T = 96^\circ\text{F}$. The numbers in this row are values of the function $G(H) = f(96, H)$, which describes how the heat index increases as the relative humidity H increases when the actual temperature is $T = 96^\circ\text{F}$. The derivative of this function when $H = 70\%$ is

the rate of change of I with respect to H when $H = 70\%$:

$$G'(70) = \lim_{h \rightarrow 0} \frac{G(70 + h) - G(70)}{h} = \lim_{h \rightarrow 0} \frac{f(96, 70 + h) - f(96, 70)}{h}$$

By taking $h = 5$ and -5 , we approximate $G'(70)$ using the tabular values:

$$G'(70) \approx \frac{G(75) - G(70)}{5} = \frac{f(96, 75) - f(96, 70)}{5} = \frac{130 - 125}{5} = 1$$

$$G'(70) \approx \frac{G(65) - G(70)}{-5} = \frac{f(96, 65) - f(96, 70)}{-5} = \frac{121 - 125}{-5} = 0.8$$

By averaging these values we get the estimate $G'(70) \approx 0.9$. This says that, when the temperature is 96°F and the relative humidity is 70% , the heat index rises about 0.9°F for every percent that the relative humidity rises.

In general, if f is a function of two variables x and y , suppose we let only x vary while keeping y fixed, say $y = b$, where b is a constant. Then we are really considering a function of a single variable x , namely, $g(x) = f(x, b)$. If g has a derivative at a , then we call it the **partial derivative of f with respect to x at (a, b)** and denote it by $f_x(a, b)$. Thus

(1)

$$f_x(a, b) = g'(a) \quad \text{where} \quad g(x) = f(x, b)$$

By the definition of a derivative, we have

$$g'(a) = \lim_{h \rightarrow 0} \frac{g(a + h) - g(a)}{h}$$

and so Equation 1 becomes

(2)

$$f_x(a, b) = \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h}$$

Similarly, the **partial derivative of f with respect to y at (a, b)** , denoted by $f_y(a, b)$, is obtained by keeping x fixed ($x = a$) and finding the ordinary derivative at b of the function $G(y) = f(a, y)$:

(3)

$$f_y(a, b) = \lim_{h \rightarrow 0} \frac{f(a, b + h) - f(a, b)}{h}$$

With this notation for partial derivatives, we can write the rates of change of the heat index I with respect to the actual temperature T and relative humidity H when $T = 96^\circ\text{F}$ and $H = 70\%$ as follows:

$$f_T(96, 70) \approx 3.75 \quad f_H(96, 70) \approx 0.9$$

If we now let the point (a, b) vary in Equations 2 and 3, f_x and f_y become functions of two variables.

(4) If f is a function of two variables, its **partial derivatives** are the functions f_x and f_y defined by

$$f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h}$$

$$f_y(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y + h) - f(x, y)}{h}$$

An alternative notation for partial derivatives is similar to Leibniz notation but uses the symbol ∂ instead of d . If $z = f(x, y)$, we write

$$f_x(x, y) = f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x, y) = \frac{\partial z}{\partial x}$$

$$f_y(x, y) = f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} f(x, y) = \frac{\partial z}{\partial y}$$

To compute partial derivatives, all we have to do is remember from Equation 1 that the partial derivative with respect to x is just the *ordinary* derivative of the function g of a single variable that we get by keeping y fixed. Thus we have the following rule.

Rule for Finding Partial Derivatives of $z = f(x, y)$

1. To find f_x , regard y as a constant and differentiate $f(x, y)$ with respect to x .
2. To find f_y , regard x as a constant and differentiate $f(x, y)$ with respect to y .

EXAMPLE 1 | If $f(x, y) = x^3 + x^2y^3 - 2y^2$, find $f_x(2, 1)$ and $f_y(2, 1)$.

SOLUTION Holding y constant and differentiating with respect to x , we get

$$f_x(x, y) = 3x^2 + 2xy^3$$

and so

$$f_x(2, 1) = 3 \cdot 2^2 + 2 \cdot 2 \cdot 1^3 = 16$$

Holding x constant and differentiating with respect to y , we get

$$f_y(x, y) = 3x^2y^2 - 4y$$

$$f_y(2, 1) = 3 \cdot 2^2 \cdot 1^2 - 4 \cdot 1 = 8$$

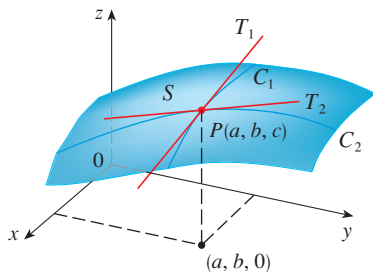


FIGURE 1

The partial derivatives of f at (a, b) are the slopes of the tangents to C_1 and C_2 .

Interpretations of Partial Derivatives

To give a geometric interpretation of partial derivatives, we recall that the equation $z = f(x, y)$ represents a surface S (the graph of f). If $f(a, b) = c$, then the point $P(a, b, c)$ lies on S . By fixing $y = b$, we are restricting our attention to the curve C_1 in which the vertical plane $y = b$ intersects S . (In other words, C_1 is the trace of S in the plane $y = b$.) Likewise, the vertical plane $x = a$ intersects S in a curve C_2 . Both of the curves C_1 and C_2 pass through the point P . (See Figure 1.)

Notice that the curve C_1 is the graph of the function $g(x) = f(x, b)$, so the slope of its tangent T_1 at P is $g'(a) = f_x(a, b)$. The curve C_2 is the graph of the function $G(y) = f(a, y)$, so the slope of its tangent T_2 at P is $G'(b) = f_y(a, b)$.

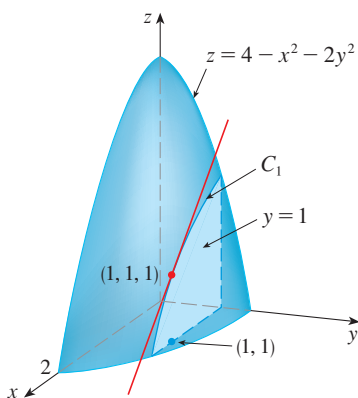


FIGURE 2

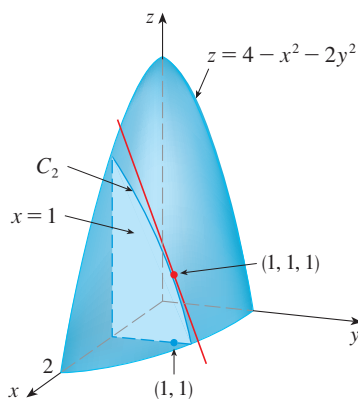


FIGURE 3

Thus the partial derivatives $f_x(a, b)$ and $f_y(a, b)$ can be interpreted geometrically as the slopes of the tangent lines at $P(a, b, c)$ to the traces C_1 and C_2 of S in the planes $y = b$ and $x = a$.

EXAMPLE 2 | If $f(x, y) = 4 - x^2 - 2y^2$, find $f_x(1, 1)$ and $f_y(1, 1)$ and interpret these numbers as slopes.

SOLUTION We have

$$\begin{aligned} f_x(x, y) &= -2x & f_y(x, y) &= -4y \\ f_x(1, 1) &= -2 & f_y(1, 1) &= -4 \end{aligned}$$

The graph of f is the paraboloid $z = 4 - x^2 - 2y^2$ and the vertical plane $y = 1$ intersects it in the parabola $z = 2 - x^2$, $y = 1$. (As in the preceding discussion, we label it C_1 in Figure 2.) The slope of the tangent line to this parabola at the point $(1, 1, 1)$ is $f_x(1, 1) = -2$. Similarly, the curve C_2 in which the plane $x = 1$ intersects the paraboloid is the parabola $z = 3 - 2y^2$, $x = 1$, and the slope of the tangent line at $(1, 1, 1)$ is $f_y(1, 1) = -4$. (See Figure 3.)

As we have seen in the case of the heat index function, partial derivatives can also be interpreted as *rates of change*. If $z = f(x, y)$, then $\partial z / \partial x$ represents the rate of change of z with respect to x when y is fixed. Similarly, $\partial z / \partial y$ represents the rate of change of z with respect to y when x is fixed.

EXAMPLE 3 | **Body mass index** In Example 9.1.2 we defined the body mass index of a person as

$$B(m, h) = \frac{m}{h^2}$$

Calculate the partial derivatives of B for a man with $m = 64$ kg and $h = 1.68$ m and interpret them.

SOLUTION Regarding h as a constant, we see that the partial derivative with respect to m is

$$\frac{\partial B}{\partial m}(m, h) = \frac{\partial}{\partial m} \left(\frac{m}{h^2} \right) = \frac{1}{h^2}$$

so

$$\frac{\partial B}{\partial m}(64, 1.68) = \frac{1}{(1.68)^2} \approx 0.35 \text{ (kg/m}^2\text{)/kg}$$

This is the rate at which his BMI increases with respect to his weight when he weighs 64 kg and his height is 1.68 m. So if his weight increases by a small amount, a kilogram for instance, and his height remains unchanged, then his BMI will increase by about 0.35 kg/m².

Now we regard m as a constant. The partial derivative with respect to h is

$$\frac{\partial B}{\partial h}(m, h) = \frac{\partial}{\partial h} \left(\frac{m}{h^2} \right) = m \left(-\frac{2}{h^3} \right) = -\frac{2m}{h^3}$$

so

$$\frac{\partial B}{\partial h}(64, 1.68) = -\frac{2 \cdot 64}{(1.68)^3} \approx -27 \text{ (kg/m}^2\text{)/m}$$

This is the rate at which his BMI increases with respect to his height when he weighs 64 kg and his height is 1.68 m. So if his weight stays unchanged while his height increases by a small amount, say 1 cm, then his BMI will *decrease* by about $27(0.01) = 0.27 \text{ kg/m}^2$.

EXAMPLE 4 | If $f(x, y) = \sin\left(\frac{x}{1+y}\right)$, calculate $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

SOLUTION Using the Chain Rule for functions of one variable, we have

$$\begin{aligned}\frac{\partial f}{\partial x} &= \cos\left(\frac{x}{1+y}\right) \cdot \frac{\partial}{\partial x}\left(\frac{x}{1+y}\right) = \cos\left(\frac{x}{1+y}\right) \cdot \frac{1}{1+y} \\ \frac{\partial f}{\partial y} &= \cos\left(\frac{x}{1+y}\right) \cdot \frac{\partial}{\partial y}\left(\frac{x}{1+y}\right) = -\cos\left(\frac{x}{1+y}\right) \cdot \frac{x}{(1+y)^2}\end{aligned}$$

EXAMPLE 5 | **BB** **Infectious disease control** In Example 9.1.12 we explored a model for the effect of vaccination and quarantine on the spread of SARS. We used the equation

$$R_0(d, v) = 5(1 - v) \frac{d}{1 + d}$$

where R_0 is the average number of new infections that each infected individual produces, v is the fraction of the population that is vaccinated, and d is the average number of days an individual remains in the population while infectious. Suppose that an outbreak has just begun and there is currently no vaccination ($v = 0$). Also suppose that there is currently no quarantine, and infectious individuals therefore circulate in the population for an average of four days. Evaluate R_0 , $\partial R_0 / \partial d$, and $\partial R_0 / \partial v$ at $d = 4$ and $v = 0$, and provide a biological interpretation of the results.

SOLUTION The value $R_0(4, 0) = 5 \cdot \frac{4}{5} = 4$ means that each infected individual is currently causing, on average, four new infections.

The partial derivative of R_0 with respect to d is

$$\frac{\partial R_0}{\partial d}(d, v) = 5(1 - v) \cdot \frac{(1 + d) \cdot 1 - d \cdot 1}{(1 + d)^2} = \frac{5(1 - v)}{(1 + d)^2}$$

so
$$\frac{\partial R_0}{\partial d}(4, 0) = \frac{5}{5^2} = \frac{1}{5}$$

This means that, if we were to introduce a small amount of quarantine (which would cause d to decrease slightly and thus reduce the spread of SARS), the value of R_0 would decline at a rate of $\frac{1}{5}$ new infections per day of decrease in d .

The partial derivative of R_0 with respect to v is

$$\frac{\partial R_0}{\partial v}(d, v) = -\frac{5d}{1 + d} \quad \text{so} \quad \frac{\partial R_0}{\partial v}(4, 0) = -\frac{5 \cdot 4}{5} = -4$$

This value means that, if we were to introduce a small amount of vaccination (which would increase v slightly and thus reduce the spread of SARS), the value of R_0 would decline at a rate of four new infections per unit increase in v . Thus the spread of SARS is more sensitive to vaccination than to quarantine.

Some computer algebra systems can plot surfaces defined by implicit equations in three variables. Figure 4 shows such a plot of the surface defined by the equation in Example 6.



FIGURE 4

EXAMPLE 6 | Find $\partial z/\partial x$ and $\partial z/\partial y$ if z is defined implicitly as a function of x and y by the equation

$$x^3 + y^3 + z^3 + 6xyz = 1$$

SOLUTION To find $\partial z/\partial x$, we differentiate implicitly with respect to x , being careful to treat y as a constant:

$$3x^2 + 3z^2 \frac{\partial z}{\partial x} + 6yz + 6xy \frac{\partial z}{\partial x} = 0$$

Solving this equation for $\partial z/\partial x$, we obtain

$$\frac{\partial z}{\partial x} = -\frac{x^2 + 2yz}{z^2 + 2xy}$$

Similarly, implicit differentiation with respect to y gives

$$\frac{\partial z}{\partial y} = -\frac{y^2 + 2xz}{z^2 + 2xy}$$

■ Functions of More Than Two Variables

Partial derivatives can also be defined for functions of three or more variables. For example, if f is a function of three variables x , y , and z , then its partial derivative with respect to x is defined as

$$f_x(x, y, z) = \lim_{h \rightarrow 0} \frac{f(x + h, y, z) - f(x, y, z)}{h}$$

and it is found by regarding y and z as constants and differentiating $f(x, y, z)$ with respect to x . If $w = f(x, y, z)$, then $f_x = \partial w/\partial x$ can be interpreted as the rate of change of w with respect to x when y and z are held fixed. But we can't interpret it geometrically because the graph of f lies in four-dimensional space.

In general, if u is a function of n variables, $u = f(x_1, x_2, \dots, x_n)$, its partial derivative with respect to the i th variable x_i is

$$\frac{\partial u}{\partial x_i} = \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_{i-1}, x_i + h, x_{i+1}, \dots, x_n) - f(x_1, \dots, x_i, \dots, x_n)}{h}$$

and we also write

$$\frac{\partial u}{\partial x_i} = \frac{\partial f}{\partial x_i} = f_{x_i} = D_i f$$

EXAMPLE 7 | Find f_x , f_y , and f_z if $f(x, y, z) = e^{xy} \ln z$.

SOLUTION Holding y and z constant and differentiating with respect to x , we have

$$f_x = ye^{xy} \ln z$$

Similarly,

$$f_y = xe^{xy} \ln z \quad \text{and} \quad f_z = \frac{e^{xy}}{z}$$

EXAMPLE 8 | **Migrating fish** In Example 9.1.13 we derived an expression for the energy expended by migrating fish:

$$E(u, v, d) = \frac{av^3d}{v - u}$$

where a is a positive constant, v is the speed of the fish relative to the water, u is the speed of the current, and d is the distance the fish swim upstream. Find the partial derivatives of E with respect to u , v , and d . What are their biological meanings?

SOLUTION Keeping v and d constant and differentiating with respect to u , we get

$$\frac{\partial E}{\partial u} = (av^3d) \left[-\frac{1}{(v-u)^2} (-1) \right] = \frac{av^3d}{(v-u)^2}$$

This represents the rate of change in energy expended that occurs from an increase in current speed while holding swimming speed and distance constant. It is positive, meaning that the energy expended increases as current speed increases.

Now we keep u and d constant and differentiate with respect to v using the Quotient Rule:

$$\frac{\partial E}{\partial v} = \frac{(v-u)(3av^2d) - (av^3d) \cdot 1}{(v-u)^2} = \frac{2av^3d - 3auv^2d}{(v-u)^2} = \frac{av^2d(2v-3u)}{(v-u)^2}$$

This represents the rate of change in energy expended that occurs from an increase in swimming speed while holding current speed and distance constant. This quantity can be positive (when $2v > 3u$) or negative (when $2v < 3u$). Thus the energy expended can increase or decrease as swimming speed increases, depending on the situation.

Finally we keep u and v constant and differentiate with respect to d :

$$\frac{\partial E}{\partial d} = \frac{av^3}{v-u}$$

This represents the rate of change in energy expended that occurs from an increase in distance while holding swimming speed and current speed constant. It is positive, meaning that the energy expended increases as the distance increases. ■

Higher Derivatives

If f is a function of two variables, then its partial derivatives f_x and f_y are also functions of two variables, so we can consider their partial derivatives $(f_x)_x$, $(f_x)_y$, $(f_y)_x$, and $(f_y)_y$, which are called the **second partial derivatives** of f . If $z = f(x, y)$, we use the following notation:

$$(f_x)_x = f_{xx} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 z}{\partial x^2}$$

$$(f_x)_y = f_{xy} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 z}{\partial y \partial x}$$

$$(f_y)_x = f_{yx} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 z}{\partial x \partial y}$$

$$(f_y)_y = f_{yy} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 z}{\partial y^2}$$

Thus the notation f_{xy} (or $\partial^2 f / \partial y \partial x$) means that we first differentiate with respect to x and then with respect to y , whereas in computing f_{yx} the order is reversed.

EXAMPLE 9 | Find the second partial derivatives of

$$f(x, y) = x^3 + x^2y^3 - 2y^2$$

SOLUTION In Example 1 we found that

$$f_x(x, y) = 3x^2 + 2xy^3 \quad f_y(x, y) = 3x^2y^2 - 4y$$

Therefore

$$\begin{aligned} f_{xx} &= \frac{\partial}{\partial x} (3x^2 + 2xy^3) = 6x + 2y^3 & f_{xy} &= \frac{\partial}{\partial y} (3x^2 + 2xy^3) = 6xy^2 \\ f_{yx} &= \frac{\partial}{\partial x} (3x^2y^2 - 4y) = 6xy^2 & f_{yy} &= \frac{\partial}{\partial y} (3x^2y^2 - 4y) = 6x^2y - 4 \end{aligned} \quad \blacksquare$$

Clairaut

Alexis Clairaut was a child prodigy in mathematics: he read l'Hospital's textbook on calculus when he was ten and presented a paper on geometry to the French Academy of Sciences when he was 13. At the age of 18, Clairaut published *Recherches sur les courbes à double courbure*, which was the first systematic treatise on three-dimensional analytic geometry and included the calculus of space curves.

Notice that $f_{xy} = f_{yx}$ in Example 9. This is not just a coincidence. It turns out that the mixed partial derivatives f_{xy} and f_{yx} are equal for most functions that one meets in practice. The following theorem, which was discovered by the French mathematician Alexis Clairaut (1713–1765), gives conditions under which we can assert that $f_{xy} = f_{yx}$. The proof is given in Appendix E.

Clairaut's Theorem Suppose f is defined on a disk D that contains the point (a, b) . If the functions f_{xy} and f_{yx} are both continuous on D , then

$$f_{xy}(a, b) = f_{yx}(a, b)$$

Partial derivatives of order 3 or higher can also be defined. For instance,

$$f_{xyy} = (f_{xy})_y = \frac{\partial}{\partial y} \left(\frac{\partial^2 f}{\partial y \partial x} \right) = \frac{\partial^3 f}{\partial y^2 \partial x}$$

and using Clairaut's Theorem it can be shown that $f_{xyy} = f_{yyx} = f_{yxy}$ if these functions are continuous.

EXAMPLE 10 | Calculate f_{xxyz} if $f(x, y, z) = \sin(3x + yz)$.

SOLUTION

$$\begin{aligned} f_x &= 3 \cos(3x + yz) \\ f_{xx} &= -9 \sin(3x + yz) \\ f_{xxy} &= -9z \cos(3x + yz) \\ f_{xxyz} &= -9 \cos(3x + yz) + 9yz \sin(3x + yz) \end{aligned} \quad \blacksquare$$

Partial Differential Equations

Partial derivatives occur in *partial differential equations* that express certain physical and biological laws. For instance, the partial differential equation

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$$

is called the **wave equation** and describes the motion of a waveform, which could be

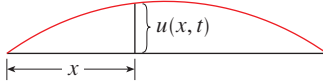


FIGURE 5

an ocean wave, a sound wave, a light wave, or a wave traveling along a vibrating string. For instance, if $u(x, t)$ represents the displacement of a vibrating violin or guitar string at time t and at a distance x from one end of the string (as in Figure 5), then $u(x, t)$ satisfies the wave equation. Here the constant a depends on the density of the string and on the tension in the string. The wave equation also arises in the description of such biological phenomena as the spread of colonies of bacteria or viruses and the propagation of nerve signals in neurons.

EXAMPLE 11 | Verify that the function $u(x, t) = \sin(x - at)$ satisfies the wave equation.

SOLUTION

$$u_x = \cos(x - at) \qquad u_t = -a \cos(x - at)$$

$$u_{xx} = -\sin(x - at) \qquad u_{tt} = -a^2 \sin(x - at) = a^2 u_{xx}$$

So u satisfies the wave equation. ■

Another partial differential equation that arises frequently in biology is the **diffusion equation**:

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2}$$

where D is a positive constant called the *diffusion constant*. This equation describes several biological processes. The function $c(x, t)$ could represent, for example, the concentration of a pollutant at time t at a distance x from the source of the pollution. The diffusion equation can also describe the invasion of alien species into a new habitat, or the diffusion of heat through a solid, or the movement of organisms along chemical gradients.

In Exercise 70 you are asked to verify that the function

$$c(x, t) = \frac{1}{\sqrt{4\pi Dt}} e^{-x^2/(4Dt)}$$

is a solution of the diffusion equation.

EXERCISES 9.2

- The temperature T (in $^{\circ}\text{C}$) at a location in the Northern Hemisphere depends on the longitude x , latitude y , and time t , so we can write $T = f(x, y, t)$. Let's measure time in hours from the beginning of January.
 - What are the meanings of the partial derivatives $\partial T/\partial x$, $\partial T/\partial y$, and $\partial T/\partial t$?
 - Honolulu has longitude 158°W and latitude 21°N . Suppose that at 9:00 AM on January 1 the wind is blowing hot air to the northeast, so the air to the west and south is warm and the air to the north and east is cooler. Would you expect $f_x(158, 21, 9)$, $f_y(158, 21, 9)$, and $f_t(158, 21, 9)$ to be positive or negative? Explain.
- At the beginning of this section we discussed the function $I = f(T, H)$, where I is the heat index, T is the temperature, and H is the relative humidity. Use Table 1 to estimate $f_T(92, 60)$ and $f_H(92, 60)$. What are the practical interpretations of these values?

- The **wind-chill index** W is the perceived temperature when the actual temperature is T and the wind speed is v , so we can write $W = f(T, v)$. The following table of values is an excerpt from Table 9.1.1.

		Wind speed (km/h)					
Actual temperature ($^{\circ}\text{C}$)	v	20	30	40	50	60	70
	T						
	-10	-18	-20	-21	-22	-23	-23
	-15	-24	-26	-27	-29	-30	-30
	-20	-30	-33	-34	-35	-36	-37
	-25	-37	-39	-41	-42	-43	-44

- Estimate the values of $f_T(-15, 30)$ and $f_v(-15, 30)$. What are the practical interpretations of these values?

- (b) In general, what can you say about the signs of $\partial W/\partial T$ and $\partial W/\partial v$?
- (c) What appears to be the value of the following limit?

$$\lim_{v \rightarrow \infty} \frac{\partial W}{\partial v}$$

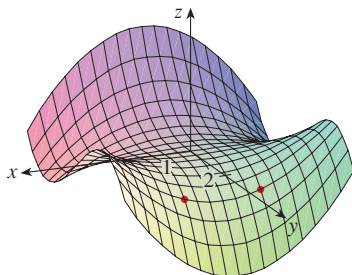
4. The wave heights h in the open sea depend on the speed v of the wind and the length of time t that the wind has been blowing at that speed. Values of the function $h = f(v, t)$ are recorded in feet in the following table.

		Duration (hours)						
Wind speed (knots)	t	5	10	15	20	30	40	50
	10	2	2	2	2	2	2	2
	15	4	4	5	5	5	5	5
	20	5	7	8	8	9	9	9
	30	9	13	16	17	18	19	19
	40	14	21	25	28	31	33	33
	50	19	29	36	40	45	48	50
	60	24	37	47	54	62	67	69

- (a) What are the meanings of the partial derivatives $\partial h/\partial v$ and $\partial h/\partial t$?
- (b) Estimate the values of $f_v(40, 15)$ and $f_t(40, 15)$. What are the practical interpretations of these values?
- (c) What appears to be the value of the following limit?

$$\lim_{t \rightarrow \infty} \frac{\partial h}{\partial t}$$

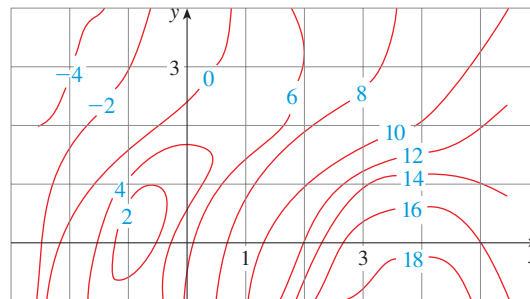
- 5–6 Determine the signs of the partial derivatives for the function f whose graph is shown.



5. (a) $f_x(1, 2)$ (b) $f_y(1, 2)$
6. (a) $f_x(-1, 2)$ (b) $f_y(-1, 2)$

7. If $f(x, y) = 16 - 4x^2 - y^2$, find $f_x(1, 2)$ and $f_y(1, 2)$ and interpret these numbers as slopes. Illustrate with either hand-drawn sketches or computer plots.

8. A contour map is given for a function f . Use it to estimate $f_x(2, 1)$ and $f_y(2, 1)$.



- 9–32 Find the first partial derivatives of the function.

9. $f(x, y) = y^5 - 3xy$ 10. $f(x, y) = x^4y^3 + 8x^2y$
11. $f(x, t) = e^{-t} \cos \pi x$ 12. $f(x, t) = \sqrt{x} \ln t$
13. $z = (2x + 3y)^{10}$ 14. $z = \tan xy$
15. $f(x, y) = \frac{x - y}{x + y}$ 16. $f(x, y) = x^y$
17. $w = \sin \alpha \cos \beta$ 18. $w = \frac{e^v}{u + v^2}$
19. $f(r, s) = r \ln(r^2 + s^2)$ 20. $f(x, t) = \arctan(x\sqrt{t})$
21. $u = te^{w/t}$ 22. $f(x, y) = \int_y^x \cos(t^2) dt$
23. $f(x, y, z) = xz - 5x^2y^3z^4$ 24. $f(x, y, z) = x \sin(y - z)$
25. $w = \ln(x + 2y + 3z)$ 26. $w = ze^{xyz}$
27. $u = xe^{-t} \sin \theta$ 28. $u = x^{y/z}$
29. $f(x, y, z, t) = xyz^2 \tan(yt)$ 30. $f(x, y, z, t) = \frac{xy^2}{t + 2z}$
31. $u = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$
32. $u = \sin(x_1 + 2x_2 + \cdots + nx_n)$

- 33–36 Find the indicated partial derivative.

33. $f(x, y) = \ln(x + \sqrt{x^2 + y^2})$; $f_x(3, 4)$
34. $f(x, y) = \arctan(y/x)$; $f_x(2, 3)$
35. $f(x, y, z) = \frac{y}{x + y + z}$; $f_y(2, 1, -1)$
36. $f(x, y, z) = \sqrt{\sin^2 x + \sin^2 y + \sin^2 z}$; $f_z(0, 0, \pi/4)$

- 37–40 Use implicit differentiation to find $\partial z/\partial x$ and $\partial z/\partial y$.

37. $x^2 + y^2 + z^2 = 3xyz$ 38. $yz = \ln(x + z)$
39. $x - z = \arctan(yz)$ 40. $\sin(xyz) = x + 2y + 3z$

- 41. Body surface area** A model for the surface area of a human body is given by the function

$$S = f(w, h) = 0.1091w^{0.425}h^{0.725}$$

where w is the weight (in pounds), h is the height (in inches), and S is measured in square feet. Calculate and interpret the partial derivatives.

(a) $\frac{\partial S}{\partial w}(160, 70)$ (b) $\frac{\partial S}{\partial h}(160, 70)$

- 42. The wind-chill index** is modeled by the function

$$W = 13.12 + 0.6215T - 11.37v^{0.16} + 0.3965Tv^{0.16}$$

where T is the temperature (in $^{\circ}\text{C}$) and v is the wind speed (in km/h). When $T = -15^{\circ}\text{C}$ and $v = 30$ km/h, by how much would you expect the apparent temperature W to drop if the actual temperature decreases by 1°C ? What if the wind speed increases by 1 km/h?

- 43. Blood flow** One of Poiseuille's laws states that the resistance of blood flowing through an artery is

$$R = C \frac{L}{r^4}$$

where L and r are the length and radius of the artery and C is a positive constant determined by the viscosity of the blood. Calculate $\partial R/\partial L$ and $\partial R/\partial r$ and interpret them.

- 44. Antibiotic concentration** If an antibiotic is administered to a patient at a constant rate through intravenous supply and is metabolized, then the concentration of antibiotic after one unit of time is

$$c(\theta, V) = c_0e^{-1/V} + \theta V(1 - e^{-1/V})$$

where c_0 is the initial concentration, θ is the rate of supply, and V is the volume of the patient's blood. Calculate $\partial c/\partial \theta$ and $\partial c/\partial V$ and interpret them.

- 45. Flapping and gliding** In the project on page 297 we expressed the power needed by a bird during its flapping mode as

$$P(v, x, m) = Av^3 + \frac{B(mg/x)^2}{v}$$

where A and B are constants specific to a species of bird, v is the velocity of the bird, m is the mass of the bird, g is the acceleration due to gravity, and x is the fraction of the flying time spent in flapping mode. Calculate $\partial P/\partial v$, $\partial P/\partial x$, and $\partial P/\partial m$ and interpret them.

- 46. Dialysis** removes urea from a patient's blood by diverting some blood flow externally through a dialyzer. The rate at which urea is removed from the blood (in mg/min) is modeled by the equation

$$c(K, t, V) = \frac{K}{V} c_0 e^{-Kt/V}$$

where K is the rate of blood flow through the dialyzer (in mL/min), t is the time (in min), V is the volume of the

patient's blood (in mL), and c_0 is the amount of urea in the blood (in mg) at time $t = 0$. Calculate $\partial c/\partial K$, $\partial c/\partial t$, and $\partial c/\partial V$ and interpret them.

- 47. Lizard energy expenditure** The average energy E (in kcal) needed for a lizard to walk or run a distance of 1 km has been modeled by the equation

$$E(m, v) = 2.65m^{0.66} + \frac{3.5m^{0.75}}{v}$$

where m is the body mass of the lizard (in grams) and v is its speed (in km/h). Calculate $E_m(400, 8)$ and $E_v(400, 8)$ and interpret your answers.

Source: Adapted from C. Robbins, *Wildlife Feeding and Nutrition*, 2nd ed. (San Diego: Academic Press, 1993).



- 48. Snake reversals and stripes** In a study of the survivorship of juvenile garter snakes, a researcher arrived at the model

$$F(R, S) = 4.2 + 0.008R + 0.102S + 0.017R^2 - 0.034S^2 - 0.268RS$$

where F is a measure of the fitness of the snake, R measures the number of reversals of direction during flight from a predator, and S measures the degree of stripedness in the color pattern of the snake. Calculate and interpret the partial derivatives.

(a) $\frac{\partial F}{\partial R}(2, 3)$ (b) $\frac{\partial F}{\partial S}(2, 3)$

Source: Adapted from E. Brodie III, "Correlational Selection for Color Pattern and Antipredator Behavior in the Garter Snake *Thamnophis ordinoides*," *Evolution* 46 (1992): 1284–98.

- 49. The van der Waals equation** for n moles of a gas is

$$\left(P + \frac{n^2a}{V^2}\right)(V - nb) = nRT$$

where P is the pressure, V is the volume, and T is the temperature of the gas. The constant R is the universal gas constant and a and b are positive constants that are characteristic of a particular gas. Calculate $\partial T/\partial P$ and $\partial P/\partial V$.

50. (a) The gas law for a fixed mass m of an ideal gas at absolute temperature T , pressure P , and volume V is $PV = mRT$, where R is the gas constant. Show that

$$\frac{\partial P}{\partial V} \frac{\partial V}{\partial T} \frac{\partial T}{\partial P} = -1$$

- (b) Show that, for an ideal gas,

$$T \frac{\partial P}{\partial T} \frac{\partial V}{\partial T} = mR$$

51–56 Find all the second partial derivatives.

51. $f(x, y) = x^3y^5 + 2x^4y$ 52. $f(x, y) = \sin^2(mx + ny)$

53. $w = \sqrt{u^2 + v^2}$ 54. $v = \frac{xy}{x - y}$

55. $z = \arctan \frac{x + y}{1 - xy}$ 56. $v = e^{xe^y}$

57–58 Verify that the conclusion of Clairaut's Theorem holds, that is, $u_{xy} = u_{yx}$.

57. $u = xe^{xy}$ 58. $u = \tan(2x + 3y)$

59–64 Find the indicated partial derivative(s).

59. $f(x, y) = 3xy^4 + x^3y^2$; f_{xxy} , f_{yyy}

60. $f(x, t) = x^2e^{-ct}$; f_{ttt} , f_{txx}

61. $f(x, y, z) = \cos(4x + 3y + 2z)$; f_{xyz} , f_{yzz}

62. $f(r, s, t) = r \ln(rs^2t^3)$; f_{rss} , f_{rst}

63. $u = e^{r\theta} \sin \theta$; $\frac{\partial^3 u}{\partial r^2 \partial \theta}$

64. $u = x^a y^b z^c$; $\frac{\partial^6 u}{\partial x \partial y^2 \partial z^3}$

65. If $f(x, y, z) = xy^2z^3 + \sec^2(x\sqrt{z})$, find f_{xzy} . [Hint: Which order of differentiation is easiest?]

66. If $g(x, y, z) = \sqrt{1 + xz} + \sqrt{1 - xy}$, find g_{xyz} . [Hint: Use a different order of differentiation for each term.]

67. Verify that the function $u = e^{-\alpha^2 k^2 t} \sin kx$ is a solution of the heat conduction equation $u_t = \alpha^2 u_{xx}$.

68. Determine whether each of the following functions is a solution of Laplace's equation $u_{xx} + u_{yy} = 0$.

(a) $u = x^2 + y^2$

(b) $u = x^2 - y^2$

(c) $u = x^3 + 3xy^2$

(d) $u = \ln \sqrt{x^2 + y^2}$

(e) $u = e^{-x} \cos y - e^{-y} \cos x$

69. Show that each of the following functions is a solution of the wave equation $u_{tt} = a^2 u_{xx}$.

(a) $u = \sin(kx) \sin(akt)$

(b) $u = t/(a^2 t^2 - x^2)$

(c) $u = (x - at)^6 + (x + at)^6$

(d) $u = \sin(x - at) + \ln(x + at)$

- 70. Diffusion equation** Verify that the function

$$c(x, t) = \frac{1}{\sqrt{4\pi Dt}} e^{-x^2/(4Dt)}$$

is a solution of the diffusion equation

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2}$$

71. If $u = xe^y + ye^x$, show that

$$\frac{\partial^3 u}{\partial x^3} + \frac{\partial^3 u}{\partial y^3} = x \frac{\partial^3 u}{\partial x \partial y^2} + y \frac{\partial^3 u}{\partial x^2 \partial y}$$

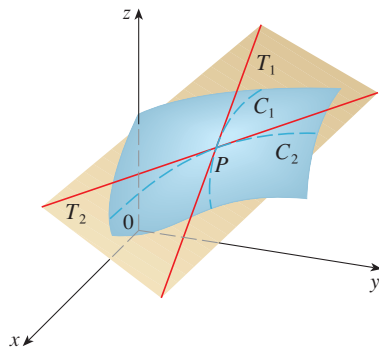
72. Show that the Cobb-Douglas production function $P = bL^\alpha K^\beta$ satisfies the equation

$$L \frac{\partial P}{\partial L} + K \frac{\partial P}{\partial K} = (\alpha + \beta)P$$

73. You are told that there is a function f whose partial derivatives are $f_x(x, y) = x + 4y$ and $f_y(x, y) = 3x - y$. Should you believe it?

9.3 Tangent Planes and Linear Approximations

One of the most important ideas in single-variable calculus is that as we zoom in toward a point on the graph of a differentiable function, the graph becomes indistinguishable from its tangent line and we can approximate the function by a linear function. (See Section 3.8.) Here we develop similar ideas in three dimensions. As we zoom in toward a point on a surface that is the graph of a differentiable function of two variables, the surface looks more and more like a plane (its tangent plane) and we can approximate the function by a linear function of two variables.

**FIGURE 1**

The tangent plane contains the tangent lines T_1 and T_2 .

Tangent Planes

Suppose a surface S has equation $z = f(x, y)$, where f has continuous first partial derivatives, and let $P(x_0, y_0, z_0)$ be a point on S . As in the preceding section, let C_1 and C_2 be the curves obtained by intersecting the vertical planes $y = y_0$ and $x = x_0$ with the surface S . Then the point P lies on both C_1 and C_2 . Let T_1 and T_2 be the tangent lines to the curves C_1 and C_2 at the point P . Then the **tangent plane** to the surface S at the point P is defined to be the plane that contains both tangent lines T_1 and T_2 . (See Figure 1.)

It can be proved that if C is any other curve that lies on the surface S and passes through P , then its tangent line at P also lies in the tangent plane. Therefore you can think of the tangent plane to S at P as consisting of all possible tangent lines at P to curves that lie on S and pass through P . The tangent plane at P is the plane that most closely approximates the surface S near the point P .

We know from Section 8.3 that any plane passing through the point $P(x_0, y_0, z_0)$ has an equation of the form

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

By dividing this equation by C and letting $a = -A/C$ and $b = -B/C$, we can write it in the form

$$(1) \quad z - z_0 = a(x - x_0) + b(y - y_0)$$

If Equation 1 represents the tangent plane at P , then its intersection with the plane $y = y_0$ must be the tangent line T_1 . Setting $y = y_0$ in Equation 1 gives

$$z - z_0 = a(x - x_0) \quad y = y_0$$

and we recognize this as the equation (in point-slope form) of a line with slope a . But from Section 9.2 we know that the slope of the tangent T_1 is $f_x(x_0, y_0)$. Therefore $a = f_x(x_0, y_0)$.

Similarly, putting $x = x_0$ in Equation 1, we get $z - z_0 = b(y - y_0)$, which must represent the tangent line T_2 , so $b = f_y(x_0, y_0)$.

Note the similarity between the equation of a tangent plane and the equation of a tangent line:

$$y - y_0 = f'(x_0)(x - x_0)$$

(2) Suppose f has continuous partial derivatives. An equation of the tangent plane to the surface $z = f(x, y)$ at the point $P(x_0, y_0, z_0)$ is

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

EXAMPLE 1 Find the tangent plane to the paraboloid $z = 2x^2 + y^2$ at the point $(1, 1, 3)$.

SOLUTION Let $f(x, y) = 2x^2 + y^2$. Then

$$f_x(x, y) = 4x \quad f_y(x, y) = 2y$$

$$f_x(1, 1) = 4 \quad f_y(1, 1) = 2$$

Then (2) gives the equation of the tangent plane at $(1, 1, 3)$ as

$$z - 3 = 4(x - 1) + 2(y - 1)$$

or

$$z = 4x + 2y - 3$$

TEC Visual 9.3 shows an animation of Figures 2 and 3.

Figure 2(a) shows the paraboloid and its tangent plane at $(1, 1, 3)$ that we found in Example 1. In parts (b) and (c) we zoom in toward the point $(1, 1, 3)$ by restricting the domain of the function $f(x, y) = 2x^2 + y^2$. Notice that the more we zoom in, the flatter the graph appears and the more it resembles its tangent plane.

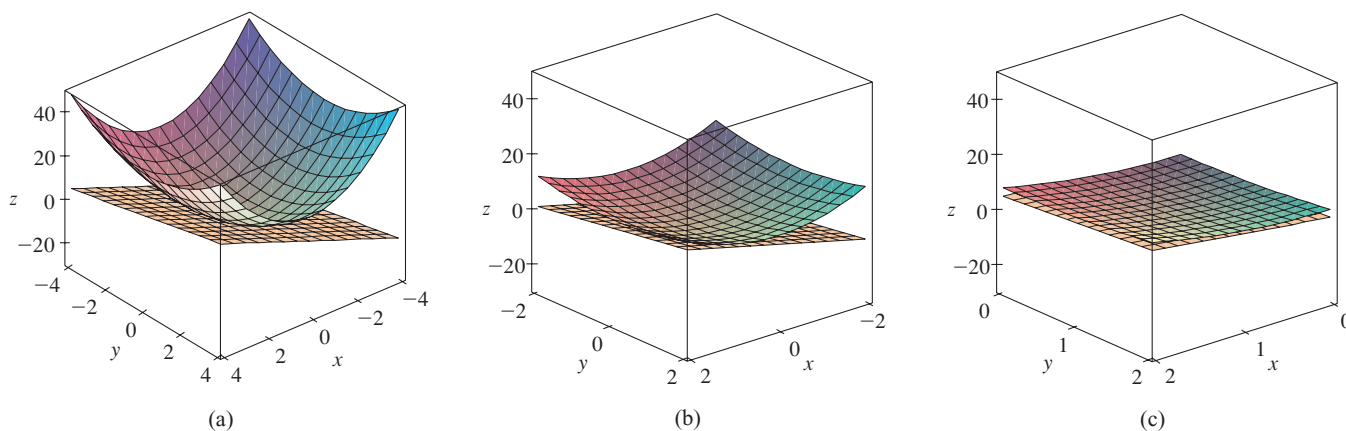


FIGURE 2 The paraboloid $z = 2x^2 + y^2$ appears to coincide with its tangent plane as we zoom in toward $(1, 1, 3)$.

In Figure 3 we corroborate this impression by zooming in toward the point $(1, 1)$ on a contour map of the function $f(x, y) = 2x^2 + y^2$. Notice that the more we zoom in, the more the level curves look like equally spaced parallel lines, which is characteristic of a plane.

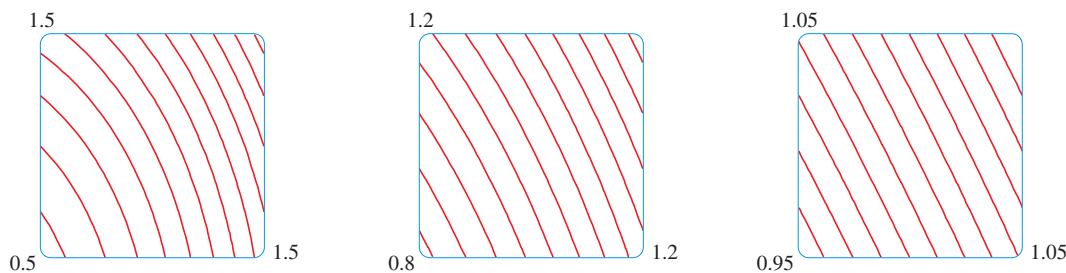


FIGURE 3 Zooming in toward $(1, 1)$ on a contour map of $f(x, y) = 2x^2 + y^2$

Linear Approximations

In Example 1 we found that an equation of the tangent plane to the graph of the function $f(x, y) = 2x^2 + y^2$ at the point $(1, 1, 3)$ is $z = 4x + 2y - 3$. Therefore, in view of the visual evidence in Figures 2 and 3, the linear function of two variables

$$L(x, y) = 4x + 2y - 3$$

is a good approximation to $f(x, y)$ when (x, y) is near $(1, 1)$. The function L is called the *linearization* of f at $(1, 1)$ and the approximation

$$f(x, y) \approx 4x + 2y - 3$$

is called the *linear approximation* or *tangent plane approximation* of f at $(1, 1)$.

For instance, at the point $(1.1, 0.95)$ the linear approximation gives

$$f(1.1, 0.95) \approx 4(1.1) + 2(0.95) - 3 = 3.3$$

which is quite close to the true value of $f(1.1, 0.95) = 2(1.1)^2 + (0.95)^2 = 3.3225$. But

if we take a point farther away from $(1, 1)$, such as $(2, 3)$, we no longer get a good approximation. In fact, $L(2, 3) = 11$ whereas $f(2, 3) = 17$.

In general, we know from (2) that an equation of the tangent plane to the graph of a function f of two variables at the point $(a, b, f(a, b))$ is

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

The linear function whose graph is this tangent plane, namely,

$$(3) \quad L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

is called the **linearization** of f at (a, b) and the approximation

$$(4) \quad f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

is called the **linear approximation** or the **tangent plane approximation** of f at (a, b) .

We have defined tangent planes for surfaces $z = f(x, y)$, where f has continuous first partial derivatives. What happens if f_x and f_y are not continuous? Figure 4 pictures such a function; its equation is

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

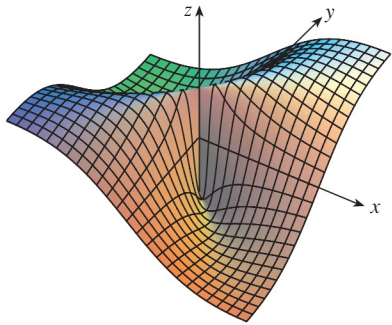


FIGURE 4

$$f(x, y) = \frac{xy}{x^2 + y^2} \text{ if } (x, y) \neq (0, 0), \\ f(0, 0) = 0$$

It can be shown that its partial derivatives exist at the origin and, in fact, $f_x(0, 0) = 0$ and $f_y(0, 0) = 0$, but f_x and f_y are not continuous. The linear approximation would be $f(x, y) \approx 0$, but $f(x, y) = \frac{1}{2}$ at all points on the line $y = x$. So a function of two variables can behave badly even though both of its partial derivatives exist. To rule out such behavior, we formulate the idea of a differentiable function of two variables by requiring the distance between $f(x, y)$ and its linear approximation $L(x, y)$ to approach 0 faster than the distance from (x, y) to (a, b) .

(5) Definition The function $f(x, y)$ is **differentiable** at (a, b) if

$$\lim_{(x, y) \rightarrow (a, b)} \frac{|f(x, y) - L(x, y)|}{\sqrt{(x - a)^2 + (y - b)^2}} = 0$$

Definition 5 says that a differentiable function is one for which the linear approximation (4) is a good approximation when (x, y) is near (a, b) . In other words, the tangent plane approximates the graph of f well near the point of tangency.

It's sometimes hard to use Definition 5 directly to check the differentiability of a function, but the next theorem (which we will not prove) provides a convenient sufficient condition for differentiability.

(6) Theorem If the partial derivatives f_x and f_y exist near (a, b) and are continuous at (a, b) , then f is differentiable at (a, b) .